# GEOMETRY BY CONSTRUCTION: A GEOMETRIC PRIOR BOTTLENECK

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## ABSTRACT

The information bottleneck (IB) principle seeks representations that retain target information while discarding input variation. The conditional entropy bottleneck (CEB) uses a label-conditional reference, but its class-wise matching terms leave the geometry between different labels unspecified. We introduce a geometric prior bottleneck (GPB) that declares this geometry statelessly: OPB uses an orthogonal class frame for classification and EPB an isometric label axis for regression. In both cases, the same geometric means are used as conditional-Gaussian KL targets and prediction anchors through a parameter-free energy or projection head. GPB preserves the CEB certificate and yields a comparator-dependent alignment inequality whose floor decomposes label ambiguity, encoder approximation, and covariance mismatch; a separate mismatch-conditioned result transfers prior separation to posterior centers. Controlled label-noise, direct mismatch, robustness, and posterior-geometry studies characterize these mechanisms, while the full benchmark comparison spans twelve dataset–backbone settings and complete IB and non-IB baselines. GPB is best or tied-best in seven settings and has a higher mean than CEB in all twelve.

## 1 INTRODUCTION

The information bottleneck (IB) principle (Tishby et al., 2000) formulates representation learning as a balance between predictive sufficiency and compression: the representation $Z$ should retain information about a target $Y$ while discarding variation in the input $X$ that does not support prediction. Variational IB (VIB) (Alemi et al., 2017) makes this trade-off trainable with a stochastic encoder and a marginal reference distribution. The conditional entropy bottleneck (CEB) (Fischer, 2020) replaces that marginal reference with a label-conditional distribution and thereby targets the residual information $I(X; Z \mid Y)$. Its variational certificate,

$$\mathbb{E}\,\mathrm{KL}(q(z|x)\|b(z|y)) \geq I(X; Z \mid Y),$$

is attractive because the reference can adapt to the label structure rather than forcing every class toward a common origin.

The same flexibility creates a geometric gap. CEB matches each posterior to the reference associated with its own label, but the certificate contains no term that compares two distinct label conditionals. Neither the marginal-reference VIB nor its variants constrain that geometry either: noise-injection bottlenecks replace the bound (Kolchinsky et al., 2019), bound comparisons clarify reference choices (Geiger & Fischer, 2020), multivariate extensions add decoders (Abdelaleem et al., 2025), and robustness deployments apply the same view at scale (Fischer & Alemi, 2020; Lee et al., 2021). Consequently, class means, directions, and pairwise margins are determined by optimization dynamics and prediction-head freedom, rather than by the compression criterion. The deterministic-label corner phenomenon identified in prior work (Kolchinsky & Tracey, 2019) is one manifestation of this freedom; a complementary issue persists even when labels are noisy or the information-plane knob moves: the certificate can be tight while the inter-label arrangement remains arbitrary. On the $T_\alpha = B_\alpha \cdot Y$ family, for example, the conditional residual can stay zero while retained label information changes across the family (Proposition 2.1; the full construction is in Appendix A.1). This observation motivates a prior whose geometry is part of the model definition.

We introduce the geometric prior bottleneck (GPB), a framework for information-bottleneck compression in which the conditional reference is a declared geometric prior rather than an unconstrained one: a stateless conditional Gaussian with bounded covariance whose means obey a two-sided declared geometry, fixed before training. The framework, not any single instance, is the object of the theory in this paper: the gap identity holds for every stateless conditional prior, and the alignment and separation results hold for any qualified geometric prior. We instantiate the framework with two concrete constructions. The orthogonal prior bottleneck (OPB) declares the classification geometry as an equal-radius orthogonal frame and uses the corresponding Gaussian anchor energies as the classifier. The equidistant prior bottleneck (EPB) declares the regression geometry as an isometric axis and uses the same axis for prediction. Thus each prior mean has two simultaneous roles: it is a target in the KL and an anchor in the readout. Both constructions are computed from the current all-label prior table, with no EMA, prototype memory, or batch-dependent geometry, and other declared geometries instantiate the same framework.

GPB reuses geometric primitives from several lines of work but attaches them to a different object. Prototype and neural-collapse methods constrain features, prototypes, or classifier weights through a classification loss or a fixed readout (Deng et al., 2014; Papyan et al., 2020; Zhu et al., 2021; Koromilas et al., 2026; Wang & Palmer, 2023); orthogonal and ETF classifiers constrain the readout (Xu et al., 2022; Ji et al., 2023); fixed frames constrain feature coordinates (Tong & Davis, 2023). OPB introduces none of these objectives: its frame is the mean map of a label-conditional Gaussian prior, and its anchor-energy readout is the Bayes rule induced by the shared covariance, so the geometric class code and the conditional KL are coupled during training. Robustness-oriented bottlenecks modify the information criterion or add adversarial objectives (Wang et al., 2021; Zhai & Zhang, 2022; Zhang et al., 2022; Hua et al., 2022); GPB keeps the ordinary CEB KL and instead transfers a declared prior geometry. Distance-preserving methods constrain deterministic feature maps (Pai et al., 2022; Luo et al., 2023), and regression compression objectives add bounds rather than geometry (Yu et al., 2024; Belucci et al., 2026); EPB places the label metric in the conditional prior, whose mean map is also the tied regression readout. Recent geometry-aware bottlenecks use the same phrase for other objects (Yan et al., 2024; Katende, 2025; Doan et al., 2026; Adilova et al., 2026). The boundary is thus at the framework level rather than the instance level: existing geometric methods are fixed constructions, while GPB is the class of IB objectives whose conditional prior carries a declared geometry, and OPB and EPB are two instances of that class evaluated in this paper. The object-level distinction is summarized in the boundary table of Appendix C.

The construction leads to two complementary theoretical statements. First, the CEB gap identity continues to hold for every stateless conditional prior. Second, for a near-optimal solution and any feasible comparator $w_0$,

$$\mathcal{D}(\widehat{w}) \le \mathcal{K}(w_0) + \frac{\mathcal{R}(w_0) + \varepsilon}{\beta},$$

so alignment is governed by a comparator's KL floor and prediction risk. The floor decomposes into irreducible label ambiguity, encoder mean approximation, and covariance mismatch (Appendix B.1). Independently, for any qualified geometric prior the mismatch-conditioned separation proposition converts the realized distance to the declared prior anchors into a lower bound on posterior-center separation. Together these results connect the constructed geometry to both optimization and the representation actually produced by the encoder.

We evaluate GPB through complementary evidence. The main table compares GPB with the complete IB and non-IB baseline set on twelve dataset–backbone settings. The controlled label-noise study follows the predicted zero-to-nonzero ambiguity floor; the direct mismatch study measures how much of the declared geometry reaches the posterior; and the compression, information-plane, robustness, and mechanism analyses examine the behavioral consequences of that transfer. Detailed uncertainty, regression curves, transfer attacks, pair-level diagnostics, and construction checks are provided in the corresponding appendices A.3–A.8.

In summary, our contributions are: (i) identifying and formalizing the missing cross-label geometric constraint in CEB; (ii) introducing GPB with OPB and EPB, whose prior means serve simultaneously as KL targets and prediction anchors; (iii) deriving comparator-dependent alignment and mismatch-conditioned separation results with an explicit ambiguity–approximation decomposition; and (iv) providing controlled and benchmark evidence that separates geometric mechanism, compression behavior, and end-to-end utility.

# 2 METHOD

## 2.1 CONSTRUCTING THE GEOMETRIC PRIOR FAMILY

Let $Y - X - Z$ and let the encoder posterior be $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \operatorname{diag}\sigma_\phi^2(x))$. CEB (Fischer, 2020) replaces the VIB marginal reference by a label-conditional reference $r(z|y)$ and uses the certificate

$$\mathbb{E}\,\mathrm{KL}(q_\phi(z|X)\|r(z|Y)) = I(X;Z|Y) + \mathbb{E}_Y\,\mathrm{KL}(q_\phi(z|Y)\|r(z|Y)) \geq I(X;Z|Y). \tag{1}$$

The certificate is a sum of class-wise matching terms. At $r = q(z|y)$ it reduces to $I(X;Z|Y)$, independently of the pairwise arrangement of the class conditionals (Proposition 2.1); the $T_\alpha$ counterexample is deferred to Appendix A.1.

**Proposition 2.1** (The certificate records nothing about inter-label geometry)**.** *The CEB compression term contains no term coupling two distinct class priors; at its class-wise matched optimum it equals*

$$\mathrm{ReX} = I(X;Z|Y), \tag{2}$$

*regardless of their inter-class means or directions.*

**Qualified geometric prior.** GPB restricts the reference to stateless conditional Gaussians

$$p_\theta(z|y) = \mathcal{N}(m_\theta(y), \Sigma_\theta(y)), \qquad \Sigma_\theta(y) \preceq \tau_{\max}^2 I, \tag{3}$$

whose means obey a declared two-sided separation for every pair of distinct discrete labels $y \neq y'$,

$$\Delta \leq \|m_\theta(y) - m_\theta(y')\| \leq \bar{\Delta}, \tag{4}$$

or a two-sided label metric condition for $P_Y$-almost every label pair $(y, y')$, where $d_Y$ is a metric on the label space $\mathcal{Y}$:

$$\rho_{\min} d_Y(y,y') \leq \|m_\theta(y) - m_\theta(y')\| \leq \rho_{\max} d_Y(y,y'), \tag{5}$$

with constants fixed before training. The objective is

$$\mathcal{L}_{\mathrm{GPB}} = \mathbb{E}[\mathrm{CE}(c_\psi(z), Y) + \beta\,\mathrm{KL}(q_\phi(z|X)\|p_\theta(z|Y))], \qquad z \sim q_\phi(z|X). \tag{6}$$

**Instances.** For classification, OPB evaluates a prior encoder on all class labels and uses its thin polar factor $Q_\theta$. If $U_\theta$ is the resulting all-class matrix, then

$$Q_\theta = U_\theta(U_\theta^\top U_\theta)^{-1/2}, \qquad Q_\theta^\top Q_\theta = I_K, \tag{7}$$

and we assume, monitoring rather than imposing it, that the monitored matrix has full column rank (Assumption A.1 in the appendix). Writing $q_{\theta,k}$ for the $k$-th column of the frame $Q_\theta$, the anchors are

$$p_\theta^{\mathrm{OPB}}(z|Y=k) = \mathcal{N}(aq_{\theta,k}, \tau^2 I), \qquad Q_\theta^\top Q_\theta = I_K, \qquad \|aq_i - aq_j\| = \sqrt{2}a. \tag{8}$$

The construction also gives the fixed geometry

$$(aq_i)^\top(aq_j) = a^2\mathbf{1}\{i=j\}, \qquad \|aq_i - aq_j\| = \sqrt{2}a \quad (i \neq j). \tag{9}$$

The anchor-energy scores are

$$s_k(z) = -\frac{\|z - aq_{\theta,k}\|^2}{2\tau^2} + \log\pi_k, \tag{10}$$

the Gaussian Bayes scores; with uniform priors this induces an orthogonal classifier whose scale is fixed by the prior (Appendix A.3). For OPB, the general objective is equivalently

$$\mathcal{L}_{\mathrm{OPB}} = \mathbb{E}\left[\mathrm{CE}(c_\psi(Z), Y) + \beta\,\mathrm{KL}\big(q_\phi(z|X)\|p_\theta^{\mathrm{OPB}}(z|Y)\big)\right], \qquad Z \sim q_\phi(\cdot|X), \tag{11}$$

so the same stateless anchors enter both the KL and the tied readout. Detailed polar construction, initialization, covariance relaxation, and the relation to prototype/ETF methods are given in Appendices A.3 and C.

For regression, EPB uses a one-dimensional continuous label, standardized once on the training set as $\tilde{y}$ with the statistics frozen. Let $W \in \mathbb{R}^{d\times 1}$ be a trainable direction with per-step normalization $u = W/\|W\|$; the isometric prior is

$$p_\theta^{\mathrm{EPB}}(z|y) = \mathcal{N}(\rho\tilde{y}\,u, \tau^2 I), \qquad \|\mu_p(y) - \mu_p(y')\| = \rho|\tilde{y} - \tilde{y}'|, \tag{12}$$

so latent distances equal $\rho$ times standardized label distances exactly. Order and magnitude of label differences are encoded with equality, which qualifies the family (Eq. equation 5 with $\rho_{\min} = \rho_{\max} = \rho$). The prediction head is the tied projection $\hat{\tilde{y}} = u^\top z/\rho$, the regression analogue of the anchor-energy classifier: no free direction or scale can bypass the prior. The main variant uses the trainable normalized axis with no bias; the fixed-axis and unconstrained-scale ablations are in Appendix A.3.

**Proposition 2.2** (Stateless GPB priors preserve the CEB certificate)**.** *For every conditional prior $p_\theta(z|y)$ independent of $x$,*

$$\mathbb{E}\,\mathrm{KL}\big(q_\phi(z|X)\|p_\theta(z|Y)\big) = I(X;Z|Y) + \mathbb{E}_Y\,\mathrm{KL}\big(q_\phi(z|Y)\|p_\theta(z|Y)\big) \geq I(X;Z|Y). \tag{13}$$

*Thus restricting the prior family changes its geometry without changing the CEB certificate identity; batch-dependent references would not satisfy this statement.*

### 2.2 Alignment and separation guarantees

Define

$$\mathcal{R}(w) = \mathbb{E}[\ell(c_\psi(Z), Y)], \qquad \mathcal{K}(w) = \mathbb{E}\,\mathrm{KL}(q_\phi(z|X)\|p_\theta(z|Y)), \tag{14}$$

and

$$\mathcal{D}(w) = \frac{\mathbb{E}\|\mu_\phi(X) - m_\theta(Y)\|^2}{2\tau_{\max}^2}. \tag{15}$$

**Theorem 2.1** (Comparator-dependent alignment under approximate realizability)**.** *If $\widehat{w}$ is $\varepsilon$-optimal for $\mathcal{R} + \beta\mathcal{K}$, then for every feasible comparator $w_0$,*

$$\mathcal{D}(\widehat{w}) \leq \mathcal{K}(\widehat{w}) \leq \mathcal{K}(w_0) + \frac{\mathcal{R}(w_0) + \varepsilon}{\beta}. \tag{16}$$

*Thus a comparator with total KL $\delta_{\mathrm{app}}$ and risk $C_{\mathrm{app}}$ gives*

$$\mathcal{D}(\widehat{w}) \leq \delta_{\mathrm{app}} + \frac{C_{\mathrm{app}} + \varepsilon}{\beta}. \tag{17}$$

*The exact $O(1/\beta)$ rate is only the zero-floor, exactly realizable special case; the full ambiguity–approximation decomposition is in Appendix A.*

For a target $d_\star$, the bound is informative only if $\mathcal{K}(w_0) < d_\star$ and

$$\beta \geq \frac{\mathcal{R}(w_0) + \varepsilon}{d_\star - \mathcal{K}(w_0)}. \tag{18}$$

A prespecified comparator's KL and risk can be estimated on an independent held-out set using

$$\widehat{\mathcal{K}}_S(w_0) = \frac{1}{|S|}\sum_{(x_i,y_i)\in S}\mathrm{KL}(q_{\phi_0}(z|x_i)\|p_{\theta_0}(z|y_i)), \quad \widehat{\mathcal{R}}_S(w_0) = \frac{1}{|S|}\sum_{(x_i,y_i)\in S}\mathbb{E}_{q_{\phi_0}}\ell(c_{\psi_0}(Z), y_i), \tag{19}$$

which estimates a comparator-specific floor, not an intrinsic dataset floor.

**Theorem 2.2** (Mismatch-conditioned posterior separation)**.** *Let*

$$D_k = \frac{\mathbb{E}[\|\mu_\phi(X) - m_\theta(k)\|^2 \mid Y = k]}{2\tau_{\max}^2}. \tag{20}$$

*For discrete labels,*

$$\|m_i - m_j\| \geq \Delta - \sqrt{2\tau_{\max}^2 D_i} - \sqrt{2\tau_{\max}^2 D_j}. \tag{21}$$

Table 1: Test accuracy (%) on classification tasks and test $R^2$ on regression tasks (mean over runs); the best entry per dataset is bolded. NCBD (classification settings only) and AdaCap (regression settings only) are external non-IB baselines; dashes mark the settings where a method does not apply.

| Dataset | Backbone | Base | VIB | SVIB | NIB | CEB | DVCCA | NCBD | AdaCap | GPB | GPB-L |
|---|---|---|---|---|---|---|---|---|---|---|---|
| MNIST | MLP | 98.33 | 98.28 | 98.15 | 98.27 | 98.43 | 98.24 | 97.95 | – | **98.54** | 98.43 |
| MNIST | CNN | 99.13 | 99.28 | 99.23 | 99.22 | 99.30 | 99.29 | 99.18 | – | **99.32** | 99.31 |
| ImageNet-100 | MLP | 83.12 | 82.84 | 83.02 | 82.35 | 82.77 | 82.99 | 83.64 | – | **84.40** | 84.01 |
| ImageNet-100 | CNN | 83.40 | 82.94 | 83.11 | 82.91 | 82.97 | 83.21 | 80.56 | – | **86.08** | 84.88 |
| Cora | GCN | 87.68 | 86.83 | 86.46 | 86.35 | 86.27 | 86.90 | 87.34 | – | **88.12** | 87.86 |
| IMDb | RNN | 84.85 | 85.50 | 85.99 | 86.14 | 85.55 | 85.69 | 86.69 | – | 86.84 | **86.99** |
| AG News | RNN | **92.94** | 92.72 | 92.86 | 92.91 | 92.83 | 92.78 | 92.37 | – | 92.87 | 92.92 |
| Cal. Housing | MLP | 0.7618 | 0.7901 | 0.7897 | 0.7862 | 0.7894 | 0.7878 | – | 0.7758 | 0.7895 | **0.7914** |
| STS-B | RNN | **0.2899** | 0.2719 | 0.2697 | 0.2417 | 0.2668 | 0.2713 | – | 0.1613 | 0.2762 | 0.2755 |
| ZINC | GCN | 0.6348 | 0.6339 | 0.6325 | 0.6178 | 0.6257 | 0.6294 | – | **0.6770** | 0.6344 | 0.6323 |
| AgeDB | MLP | 0.2180 | 0.3173 | 0.3385 | 0.2710 | 0.3317 | 0.3180 | – | 0.3482 | **0.3536** | 0.3320 |
| AgeDB | CNN | 0.3914 | 0.3342 | 0.3323 | 0.3337 | 0.3882 | 0.3205 | – | 0.4684 | **0.4700** | 0.4388 |
| Mean rank | | 4.83 | 5.75 | 5.50 | 7.00 | 5.96 | 6.00 | 6.43 | 4.40 | 1.83 | 2.54 |
| Best or tied-best | | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 7 | 2 |

*For continuous labels define*

$$D(y) = \frac{\mathbb{E}[\|\mu_\phi(X) - m_\theta(y)\|^2 \mid Y = y]}{2\tau_{\max}^2}, \tag{22}$$

*and the analogous lower bound is*

$$\|m(y) - m(y')\| \geq \rho_{\min} d_Y(y, y') - \sqrt{2\tau_{\max}^2 D(y)} - \sqrt{2\tau_{\max}^2 D(y')}, \tag{23}$$

*for almost every pair. The discrete bound is positive only when*

$$\Gamma_{ij} := \frac{\sqrt{2\tau_{\max}^2 D_i} + \sqrt{2\tau_{\max}^2 D_j}}{\|m_\theta(i) - m_\theta(j)\|} < 1; \tag{24}$$

*for OPB with equal mismatch $D$ this is $D < a^2/(4\tau_{\max}^2)$. The scale is a validation-selected design parameter and does not itself guarantee a positive bound.*

## 3 Experiments

The evaluation has three tracks: predictive comparison, controlled theory illustration, and structural diagnostics. To keep the main paper readable, the main figures show one classification compression sweep, targeted white-box attacks, and the aggregate geometry/mismatch summaries. Regression compression curves, transfer attacks, pair-level mismatch plots, and full uncertainty tables are reported in Appendices B.3, B.5, A.6, and A.7–A.8; the remaining ablations and estimator panels are collected in Appendix B.

### 3.1 Experimental setup

We evaluate twelve dataset–backbone settings over nine public datasets: MNIST, ImageNet-100, Cora, IMDb, AG News, California Housing, STS-B, ZINC-12k, and AgeDB, covering MLP, CNN, GCN, and RNN backbones. All methods use the same splits, backbone, stochastic Gaussian posterior, bottleneck dimension $d = 256$, Adam training, early stopping, and validation-based selection. The baselines are Base, VIB, SVIB, NIB, CEB, supervised $\beta$-DVCCA, and the external NCBD/AdaCap comparisons; GPB is OPB on classification rows and EPB on regression rows, with GPB-L denoting the identical prior and a free head. Full data preprocessing, grids, seeds, and baseline definitions are in Appendix A.3.

### 3.2 Main predictive comparison

Table 1 reports the best validation-selected configuration of every baseline and GPB variant; the complete standard deviations, paired confidence intervals, and per-method configuration details are

| $r$ | $\delta_{\text{noise}}$ | $\widehat{D}(\beta{=}10^{-3})$ | $\widehat{D}(\beta{=}10)$ | max excess | $\widehat{L} - L_{\text{comp}}$ |
|---|---|---|---|---|---|
| 0 | 0.000 | $0.0000 \pm 0.0000$ | $0.0000 \pm 0.0000$ | 0.000 | $0.000 \pm 0.000$ |
| 0.1 | 3.400 | $3.4142 \pm 0.0002$ | $3.4018 \pm 0.0000$ | 0.014 | $-0.005 \pm 0.004$ |
| 0.2 | 6.400 | $6.4225 \pm 0.0005$ | $6.4067 \pm 0.0002$ | 0.023 | $-0.053 \pm 0.008$ |
| 0.4 | 11.200 | $11.2324 \pm 0.0016$ | $11.2172 \pm 0.0003$ | 0.032 | $-0.191 \pm 0.016$ |

Table 2: Controlled label-noise floor (mean $\pm$ std over five seeds); the last column is the fitted objective minus the explicit comparator objective at $\beta = 10$.

in Appendices A.7 and A.8. The ranking summary is dominated by the two GPB heads: GPB has the lowest mean rank (1.83 versus 2.54 for GPB-L and 4.83 for the best plain baseline) and is best or tied-best in $7/12$ settings. The gains are largest where the task has margin to resolve them: on ImageNet-100 (CNN) GPB leads CEB by 3.11 points and on Cora by 1.85, while the saturated settings are flat (MNIST CNN and AG News, where all methods sit within 0.6 points of the best) and the small-data regressions are essentially tied with the best baselines (Housing within 0.002 $R^2$ of VIB; on STS-B the plain backbone remains best). The tied head costs nothing anywhere except where it helps: it is within 0.4 points of GPB-L in $11/12$ settings, and on ImageNet-100 (CNN) it is 1.20 points better, the setting where the free head has the most capacity to overfit. Among the external non-IB baselines, AdaCap is competitive on regression (best on ZINC), while NCBD's mean rank (6.43) places it behind the plain backbone on classification. Across modalities, the same construction improves the operating point where improvement is available and costs nothing elsewhere.

### 3.3 Controlled label ambiguity

This experiment isolates the comparator floor by making the label-ambiguity term of the alignment bound analytically known, and verifies three claims: the noise-free case realizes the zero floor exactly, the measured mismatch tracks the analytic floor at every noise level, and the fitted objective never leaves the explicit comparator's reachable set. The construction uses $K = 10$, $X = e_C$, and symmetric label noise $r \in \{0, 0.1, 0.2, 0.4\}$ with OPB $(a, \tau) = (6, 1)$. The analytic ambiguity floor is

$$\delta_{\text{noise}}(r) = \frac{a^2}{2\tau^2}\left[1 - (1-r)^2 - \frac{r^2}{K-1}\right], \tag{25}$$

which grows from 0 to 11.2 nats as $r$ runs over the noise grid. The explicit comparator takes posterior mean $h_0(x) = a \sum_k \eta_k(x) q_k$ and covariance $\tau^2 I$, where $\eta_k(x) = P(Y = k \mid X = x)$; its KL equals $\delta_{\text{noise}}(r)$ exactly, and its anchor-energy risk is $C_{\text{comp}}(r)$. The training protocol (population-weighted objective, warm start, Monte Carlo risk estimation) is in Appendix A.5. Whenever the fitted objective satisfies $\widehat{L} \le L_{\text{comp}} = C_{\text{comp}} + \beta\delta_{\text{noise}}$, non-negativity of the fitted risk gives the directly checkable comparison

$$\widehat{D} \;\le\; \widehat{\mathcal{K}} \;\le\; \delta_{\text{noise}} + \frac{C_{\text{comp}}}{\beta}. \tag{26}$$

Thus the known-noise experiment tracks the predicted floor, while the sampled-label sensitivity and comparator-gate checks are deferred to Appendix A.5.

Table 2 reads the transition from the zero to the non-zero floor column by column. The zero-noise row reproduces the exact-realizability limit to the stored precision. For $r > 0$ the measured mismatch tracks the analytic floor at both ends of the $\beta$ grid: at $\beta = 10$ it matches the floor to within 0.02 nats at every noise level, while at the smallest $\beta$ it sits above the floor by an excess that grows with the noise level (from 0.014 to 0.032 nats) and declines monotonically with compression, without implying an exact $1/\beta$ law. The last column closes the loop: the fitted objective satisfies $\widehat{L} \le L_{\text{comp}}$ at every reported point, and the margin over the explicit comparator grows with the noise level (from $-0.005$ to $-0.191$), so the trained posterior moves farther inside the comparator's reachable set exactly where the floor is largest. Sampled-label, fixed-variance, and trainable-frame sensitivity controls are deferred to Appendix A.5.

### 3.4 Direct mismatch analysis

This section pursues two checks on the learned posterior: how much of the declared anchor geometry it inherits, and whether the separation guarantee survives under the measured mismatch. For

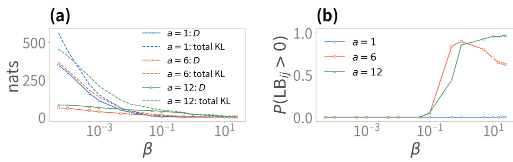


Figure 1: Direct mismatch diagnostics over the ImageNet-100 compression sweep. (a) Anchor mismatch $\widehat{D}$ (solid) and total KL (dashed) against $\beta$, by anchor scale. (b) Fraction of class pairs with a positive separation lower bound. Means are over five seeds; no checkpoint is omitted.

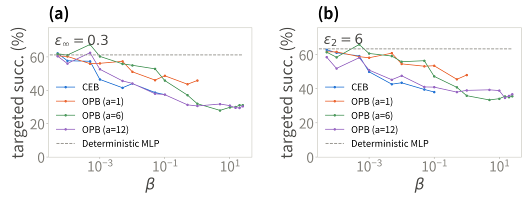


Figure 2: Targeted white-box PGD on MNIST (MLP). (a) $L_\infty$, $\varepsilon = 0.3$; (b) $L_2$, $\varepsilon = 6$. Curves use the stochastic path and are shown only where clean accuracy exceeds 90%. Transfer curves are in Appendix B.5.

each checkpoint, the empirical variance bound $\widehat{\tau}^2_{\max}$ is the largest conditional-reference variance realized over the class table (classification) or test labels (regression); all statistics are first computed within each seed and then aggregated across the five seeds, and class or bin pairs are not treated as independent replicates. For test examples of class $k$, define

$$\widehat{D}_k := \frac{1}{n_k} \sum_{n:y_n=k} \frac{\|\mu_\phi(x_n) - a q_{\theta,k}\|^2}{2\widehat{\tau}^2_{\max}}, \qquad \widehat{m}_k := \frac{1}{n_k} \sum_{n:y_n=k} \mu_\phi(x_n). \tag{27}$$

Jensen gives the per-class check $\|\widehat{m}_k - a q_{\theta,k}\| \le \sqrt{2\widehat{\tau}^2_{\max}\widehat{D}_k}$, and the triangle inequality gives the empirical separation bound

$$\widehat{\mathrm{LB}}_{ij} = \sqrt{2}a - \sqrt{2\widehat{\tau}^2_{\max}\widehat{D}_i} - \sqrt{2\widehat{\tau}^2_{\max}\widehat{D}_j}, \tag{28}$$

evaluated without clipping. Across all OPB scales, $\beta$ values, seeds, and classes, no Jensen or separation-slack violation exceeds $10^{-6}$. CEB's mismatch is shown descriptively, but its bound and slack entries are left empty because its reference has no declared $\Delta$.

On ImageNet-100, the two panels read the condition and its certified consequence together (Fig. 1). Mismatch falls with compression: from 62.3 nats at $\beta = 5 \cdot 10^{-5}$ to 2.54 at $\beta = 25$ for $a = 6$, with the total KL co-plotted, and the ordering across scales reverses over the same range. At weak compression the smallest frame is farthest from its anchors (349.8, against 82.1 and 62.3), while at $\beta = 25$ the $a = 12$ frame is tightest among the separated frames (0.56 against 2.54): the anchor-energy readout's prediction optimum sits exactly at the anchors, so a larger declared scale pulls harder and the tied readout is self-enforcing. The positive-bound fraction rises exactly where the corrections shrink: $a = 12$ climbs to 96.9% at $\beta = 25$, $a = 6$ peaks at 89.5% near $\beta = 1$ and settles at 62.3%, and $a = 1$ never certifies, its corrections exceeding the declared margin at every $\beta$ as its centers collapse under strong compression. The bound and slack readings, the decoupling scatter, and the regression panels are collected in Appendix A.6.

### 3.5 Adversarial robustness

We use the targeted MNIST PGD protocol of CEB with EOT-10 gradients and 10-sample stochastic predictions; each curve is shown only where the stochastic clean accuracy exceeds 90% (Fig. 2). Within that valid range, targeted success falls monotonically with compression: every curve starts at the deterministic baseline's level ($\approx 61\%$ under $L_\infty$, $\approx 63\%$ under $L_2$) and declines by roughly a third to a half by the end of its valid range. The models differ in how far that range extends. CEB's curve ends at $\beta = 0.1$, where its success (37.4%/38.1%) is matched by OPB $a = 12$ (37.4%/41.0%); beyond it CEB collapses toward chance accuracy, where near-zero success is trivial rather than a robustness gain, so no later point is plotted. OPB $a = 6, 12$ stay valid over the whole grid and continue to fall, reaching 31.1%/30.5% under $L_\infty$ and 35.7%/36.8% under $L_2$ at $\beta = 25$, roughly 30 points below the deterministic baseline under $L_\infty$ and about 27 under $L_2$, with clean accuracy still $\approx 98\%$; the curves plateau from $\beta \ge 5$ onward, so the gain sits in the compressed operating

| variant | MNIST (Acc %) | ImageNet-100 (Acc %) | Housing ($R^2$) |
|---|---|---|---|
| Base | $98.3_{\pm 0.1}$ | $83.1_{\pm 0.1}$ | $0.762_{\pm 0.024}$ |
| NCM-learn | $98.2_{\pm 0.2}$ | $82.2_{\pm 0.4}$ | – |
| NCM-ortho | $98.1_{\pm 0.1}$ ($a = 1$) | $82.3_{\pm 0.6}$ ($a = 1$) | – |
| CEB | $98.4_{\pm 0.1}$ ($\beta = 0.01$) | $82.8_{\pm 0.2}$ ($\beta = 0.01$) | $0.789_{\pm 0.002}$ ($\beta = 0.0001$) |
| CEB-energy | $98.2_{\pm 0.1}$ ($\beta = 0.001$) | $81.6_{\pm 0.2}$ ($\beta = 0.001$) | – |
| GPB-L | $98.4_{\pm 0.1}$ ($\beta = 0.1, a = 12$) | $83.4_{\pm 0.7}$ ($\beta = 1, a = 12$) | – |
| GPB | $98.4_{\pm 0.1}$ ($\beta = 0.1, a = 12$) | $84.2_{\pm 0.1}$ ($\beta = 1, a = 12$) | – |
| OPB-FS | $98.5_{\pm 0.0}$ ($\beta = 0.1, a = 12$) | $84.3_{\pm 0.2}$ ($\beta = 1, a = 12$) | – |
| CEB-tied | – | – | $0.781_{\pm 0.006}$ ($\beta = 0.01$) |
| EPB-L | – | – | $0.791_{\pm 0.003}$ ($\beta = 0.0001, a = 6$) |
| EPB | – | – | $0.788_{\pm 0.004}$ ($\beta = 0.0001, a = 6$) |
| EPB-FS | – | – | $0.787_{\pm 0.008}$ ($\beta = 0.001, a = 6$) |

Table 3: Attribution ablation: best-configuration test metrics (mean ± std over five seeds). Configurations are selected by the mean test metric, applied identically to every variant, so all comparisons remain paired by seed; the selection is slightly optimistic in level but shared across the table. One OPB-FS configuration (ImageNet-100, $\beta = 10^{-4}$, $a = 1$) diverged during training (free scales grow without penalty at negligible $\beta$) and is excluded.

point rather than in $\beta$ alone. OPB $a = 1$ stops at $\beta = 1$, where its sampled-path accuracy drops below $90\%$ as the posterior variance relaxes to the prior level (the same effect as in Section 3.8). Transfer attacks are reserved for the gradient-masking check in Appendix B.5.

### 3.6 Factor-wise attribution

The main comparison changes several factors simultaneously, so we also report a paired intervention ladder in a common operating environment: every variant is re-trained at its best shared configuration on the three settings of the mechanism analysis (Table 3), and all comparisons are paired by seed. On ImageNet-100, GPB reaches $84.2$ points, within $0.1$ of the free-scale OPB-FS and well above CEB ($82.8$); on MNIST all variants tie within $0.4$ points, and on Housing every EPB variant ties within $0.01$ $R^2$, consistent with the flat strong-compression plateaus of Section 3.7.

The ladder switches each factor independently. *NCM-learn* removes the bottleneck entirely: deterministic $z = \mu_\phi(x)$, energy readout on trainable prototypes, no posterior and no KL. *NCM-ortho* fixes the prototypes to the orthogonal frame $a\, e_k$: fixed geometry, still no IB. *CEB-energy* and *CEB-tied* mount the same energy readout (resp. tied projection) on CEB's trainable free reference. *OPB-FS* and *EPB-FS* keep the orthogonal frame and the energy (resp. tied) readout but make the anchor radius learnable, so only the fixed scale is switched off. Together with Base, CEB, GPB-L and GPB this isolates geometry (CEB-energy vs. OPB-FS), scale (OPB-FS vs. GPB), readout (GPB-L vs. GPB; CEB vs. CEB-energy), and the IB term itself (NCM-ortho vs. GPB).

The paired contrasts (accuracy points on ImageNet-100, $95\%$ CIs in Appendix B.2) rank the factors: geometry is the largest ($+2.67$, OPB-FS over CEB-energy), followed by the IB term ($+1.89$, GPB over NCM-ortho, which additionally contains frame trainability) and the tied readout ($+0.82$, GPB over GPB-L), while fixing the anchor scale costs essentially nothing ($-0.04$, GPB over OPB-FS). Two patterns matter as much as the ranking. First, the readout contrast flips sign with the accompanying geometry: on the free-geometry side the energy readout costs $1.19$ points (CEB-energy below CEB), so restricting the readout helps only when the geometry it ties to is the constructed one. Second, the prototype readout alone does not work: NCM-learn trails the plain backbone by $0.96$ points, so the energy form is not the source of the gain either. On MNIST all contrasts shrink to within $0.35$ points, and on Housing to within $0.01$ $R^2$, so the ladder resolves factors only where the task margin is large enough. These interventions change the fitted training problem and therefore support factor-wise attribution, not a unique causal decomposition.

### 3.7 Compression diagnostics

For compactness, the main text shows the classification sweep on ImageNet-100 (MLP); the companion California Housing regression sweep is in Appendix B.3. The larger OPB anchors remain on a high-accuracy plateau farther into strong compression, while CEB and $a = 1$ can collapse; larger anchors also incur a weak-compression cost. The curves make the anchor-scale/compression trade-off visible across the useful operating range.

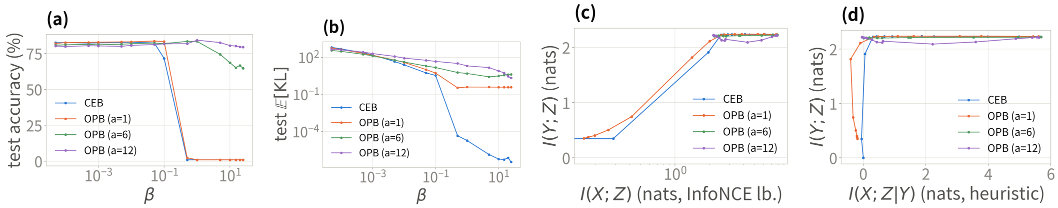


Figure 3: Compression diagnostics across two views and two tasks. (a)–(b) ImageNet-100 classification sweep (MLP): test accuracy and realized conditional KL against $\beta$; CEB and the smallest anchor collapse at strong compression, whereas $a = 12$ retains the plateau. The Housing regression counterpart is provided in Appendix B.3. (c)–(d) MNIST information-plane and certificate-plane estimates (estimator-level diagnostics rather than mutual-information identities; Appendix B.4).

At $\beta = 25$, OPB with $a = 12$ remains at $79.4\%$ accuracy while CEB has already left its valid high-accuracy range; at weak compression the fixed-scale head pays a one-to-three point cost. The realized KL panel shows where the two objectives spend their penalty: CEB's KL collapses as its learned reference shrinks to absorb the term (3.6 at $\beta = 0.1$ to $\sim 10^{-5}$ at $\beta = 0.5$), whereas OPB's KL stays anchored at the geometric scale of its fixed prior ($a = 12$: 48 at $\beta = 0.1$, 21 at $\beta = 1$) and is dominated by the mean-mismatch term, so the anchor structure is retained rather than absorbed.

### 3.8 Information-plane diagnostic

The MNIST information-plane estimates provide a second view of the compression behavior (Fig. 3c,d). They are estimator-level diagnostics rather than mutual-information identities; the matched table and estimator details are in Appendix B.4, with the Housing companion in Appendix A.9.

The two planes show what each method discards as compression deepens. In the information plane (Fig. 3c, log $I(X;Z)$ axis), CEB follows the classic corner: compression and task information fall together, from $I(X;Z) = 2.53$ nat at $\beta = 0.1$ to 1.96 nat at $\beta = 0.5$, and by $\beta = 1$ both are near zero. OPB with $a = 6, 12$ compresses from $\approx 7.9$ to $\approx 2.2$ nat but never below it, with $I(Y;Z)$ flat at the $H(Y) = \ln 10 \approx 2.30$ ceiling, so compression stops at the task-information level. $a = 1$ tracks $a = 6, 12$ until $\beta = 0.5$ and then collapses toward CEB's corner, with $I(Y;Z)$ falling to 0.35–0.74 nat (Appendix B.4). The certificate plane (Fig. 3d) reads the same contrast as where each trajectory ends: CEB crosses the zero-residual line and $I(Y;Z)$ collapses with it, whereas $a = 6, 12$ terminate on it within $\pm 0.2$ nat of zero residual at full task height. Negative residuals are estimation error rather than information.

### 3.9 Structural diagnostics

At matched compression $(\beta, a, \rho) = (0.1, 12, 12)$, the posterior geometry is read directly off the class-center cosine Gram matrices (Fig. 4a,b): OPB's Gram is visually an identity matrix, with off-diagonal mean $0.0203$, whereas CEB's shows visible off-diagonal structure ($0.1205$) with no clean pattern. The posterior thus inherits the constructed orthogonal frame, which is exactly what the zero-parameter tied readout relies on; CEB's free reference leaves its posterior entangled, and its free head must re-absorb that geometry. The tied heads attain $98.44\%$ on MNIST against CEB's $98.16\%$, and $R^2 = 0.778$ on Housing against $0.491$. On Housing (Fig. 4c,d), the axial scatter shows the same contrast: EPB's points hug its prior line (off-axis fraction $0.0002$), whereas CEB's cloud deviates from its prior line's direction ($0.0233$) and its learned axis scale is $2.65 \pm 0.59$ across runs, so its geometry is not a stable structure for a tied readout to exploit.

**Conclusion and limitations.** The experiments form a consistent progression from construction to behavior: the baseline table establishes end-to-end utility across modalities; the factor interventions attribute the gain to the geometric prior; the controlled noise study matches the analytic ambiguity floor; and the mismatch, robustness, compression, information-plane, and geometry diagnostics show how the declared prior is reflected in the learned posterior at different operating points. Constraining the conditional prior changes the geometry available to the KL, and the tied readout makes

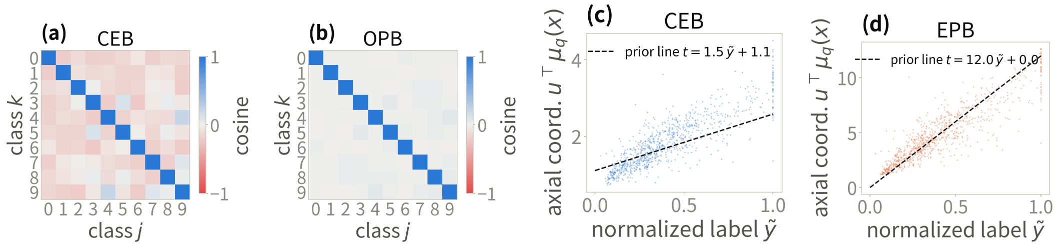


Figure 4: Posterior geometry at the matched operating point: class-center Gram matrices for (a) CEB and (b) OPB, and axial coordinates for (c) CEB and (d) EPB.

| Model | Geometry concentration | Scale following | Tied metric | Cross-seed stability |
|---|---|---|---|---|
| CEB (MNIST) | $0.1205 \pm 0.0026$ | 5.14 | 98.16% | 0.0201 |
| OPB (MNIST) | $0.0203 \pm 0.0021$ | 11.00 | 98.44% | 0.0082 |
| CEB (Housing) | 0.0233 | $2.65 \pm 0.59$ | $R^2 = 0.491$ | 0.593 |
| EPB (Housing) | 0.0002 | $9.22 \pm 0.15$ | $R^2 = 0.778$ | 0.155 |

Table 4: Compact mechanism snapshot at matched compression (cross-seed stability is the standard deviation over the five runs of the geometric concentration quantity). Full definitions are in the stored evaluation report.

that geometry visible to prediction; the measured mismatch determines how much of the declared geometry transfers, and the anchor scale determines where the trade-off is useful.

The results are qualified accordingly. The alignment bound is comparator-relative: on a real benchmark its floor is determined by a prespecified comparator and its label, mean-approximation, and covariance terms rather than an intrinsic dataset constant, and the separation lower bound is informative only when the realized mismatch stays below the declared margin, so anchor scale remains an operating choice. The information-plane quantities are estimator-level diagnostics rather than mutual-information identities. Data-driven comparator certificates, adaptive scale selection, and tighter information estimates are natural next steps. Proofs and all supplementary evidence are collected in Appendices A–C.

## AI USE STATEMENT

(This section is **required** and does not count toward the page limit.)

In this work, we used generative AI tools for drafting and editing portions of the code and of this manuscript. All AI-assisted content was manually reviewed and verified by the authors; experimental results and theoretical claims were produced and checked by our own code and analysis. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

## REPRODUCIBILITY STATEMENT

The code, hyperparameter grids, and evaluation protocol are described in Section 3.1; the geometric prior is stateless (no EMA or prototypes) and the orthogonalization is exact wherever defined (Assumption A.1). All datasets are public, with preprocessing steps given in the release. Proofs of all statements are collected in the appendix. Statistical claims use the paired bootstrap confidence intervals and non-inferiority margin described in Appendix A.8.

## ACKNOWLEDGMENTS

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# A Deferred proofs, construction details, and scope

## A.1 Failure diagnosis

**Proposition A.1** (Dead knob, following Caveat 1 of Kolchinsky & Tracey (2019))**.** *Assume $Y = f(X)$. For every $\gamma > 0$, the CEB objective $L_\gamma(Z) := I(X;Z|Y) - \gamma I(Y;Z)$ is minimized exactly on the corner family $\{Z : I(Y;Z) = H(Y),\ I(X;Z|Y) = 0\}$, and the map $\gamma \mapsto \big(I(X;Z^*|Y),\, I(Y;Z^*)\big)$ is constant. Consequently no interior point of the IB curve is ever a Lagrangian minimizer of CEB, over the entire parameter range.*

*Proof sketch (see Kolchinsky & Tracey (2019) for the full argument).* By the chain rule, $L_\gamma(Z) = I(X;Z) - (1+\gamma)I(Y;Z)$; the data-processing inequality gives $L_\gamma \geq -\gamma H(Y)$, attained exactly at the corner $Z = Y$. In the Caveats paper's maximizing convention this is their Eq. 7 with $\beta' = 1/(1+\gamma) \in (0,1)$, the range in which every maximizer is pinned at the copy corner. ☐

The corner is a single point of the information plane and an entire equivalence class of representations; the objective is exactly indifferent within the class, and neither the objective nor the certificate records which member the optimizer reaches. In the deterministic scenario this corner *is* the MNI point, so CEB's declared target and the degenerate attractor coincide. The Caveats paper repairs this by squaring the compression term; our anchored reference supplies the same strict convexity by a different mechanism, without squaring: a quadratic displacement term built into the KL by construction (Proposition A.5 below). We present this as a mechanism connection to the Caveats fix, not as a new solution of the deterministic degeneration.

*Proof of Proposition 2.1.* For any matched pair ($b = q(z|y)$), the gap identity of Eq. equation 1 gives Eq. equation 2; the matching condition is stated per class and no term of the certificate couples two distinct class priors, so the certificate value carries no information about the pairwise geometry of $\{q(z|y)\}$. Note that $I(X;Z|Y)$ is generally non-zero; it vanishes only under the conditional independence $Z \perp X \mid Y$, a property of the encoder, not of the prior. ☐

**Proposition A.2** (Retention blindness on $T_\alpha$)**.** *On the manifold $T_\alpha = B_\alpha \cdot Y$ of Kolchinsky & Tracey (2019) (their Eq. 6), a family that depends on $Y$ alone, $I(X;T_\alpha|Y) = 0$ for every $\alpha$, and with the optimal backward encoder and classifier,* $\mathrm{ReX}(T_\alpha) = 0$ *and* $\mathrm{CE}(T_\alpha) = H(Y) - I(Y;T_\alpha) =: \delta_\alpha$*: the certificate reads $0$ from the total collapse ($\alpha = 0$) to the full copy ($\alpha = 1$), while the retained label information $I(Y;T_\alpha)$ sweeps the whole axis. Tightness of the certificate is therefore compatible with retaining all, part, or none of $Y$: the certificate records nothing about how much label information the code retains.*

*Proof.* $T_\alpha$ is a function of $Y$ alone, hence $T_\alpha \perp X \mid Y$ and $I(X;T_\alpha|Y) = 0$. By Proposition 2.1 with the true backward encoder the certificate is tight, $\mathrm{ReX} = I(X;T_\alpha|Y) = 0$, and the optimal classifier attains $\langle \log c(y|z) \rangle = -H(Y|T_\alpha)$, i.e. $\mathrm{CE} = H(Y|T_\alpha) = \delta_\alpha$. ☐

*Remark* A.1 (The objective on $T_\alpha$ is $\beta$-independent and strictly prefers the full copy). On $T_\alpha$ the compression term contributes nothing ($\mathrm{ReX} \equiv 0$), so the objective reads $L = \mathrm{CE} + \beta\,\mathrm{ReX} = \delta_\alpha$, independent of $\beta$ and strictly minimized at $\alpha = 1$ (the full copy). No certified residual is produced anywhere on the manifold; the blindness is that the certificate cannot tell the optimizer — or the analyst — where on the retention axis the code sits.

*Scope of the comparison argument.* Theorem 2.1 applies to arbitrary $P_{XY}$ and every feasible comparator in the fitted model class; it does not require $q(z|x) = p_\theta(z|y)$. Its usefulness depends on the comparator KL. Label ambiguity, limited encoder capacity, and covariance misspecification produce the non-zero floor made explicit in Proposition B.1. The stronger statement that this floor is zero still requires the assumptions of Corollary B.1: deterministic labels, exact posterior–prior matching, and a compatible head. We do not assume those exact-realizability premises for the real-data benchmarks. Their performance comparison evaluates utility, while their structural diagnostics report realized mismatch and geometry. The controlled known-noise experiment instead evaluates the ambiguity term where its population value is analytic, including the deterministic distribution as its zero-floor special case.

### A.2 PROOFS OF THE ALIGNMENT GUARANTEES

*Proof of Proposition 2.2.* For fixed $\theta$, apply the gap identity of Eq. equation 1 to the conditional distribution $p_\theta(z|y)$ and the posterior $q_\phi$. $\square$

*Proof of Theorem 2.2.* For (a), Jensen gives $\|m_k - m_\theta(k)\| \le \sqrt{2\tau_{\max}^2 D_k}$, and the triangle inequality on $m_i - m_j = (m_i - m_\theta(i)) + (m_\theta(i) - m_\theta(j)) + (m_\theta(j) - m_j)$ gives Eq. equation 21. For (b), the same two inequalities hold pointwise for $P_Y$-a.e. $y$: Jensen on the regular conditional expectation gives $\|m(y) - m_\theta(y)\| \le \sqrt{2\tau_{\max}^2 D(y)}$ (the norm is convex), and the triangle inequality gives Eq. equation 23; intersecting the two measure-one sets of valid $y$'s yields the almost-everywhere pair statement. $\square$

*Remark* A.2 (Finite-sample estimation of the continuous case). On finite samples the regular conditional expectations of part (b) are estimated by binning the label axis into intervals $B_i$ with representative labels $\bar{y}_i$ and centers $m_i = \mathbb{E}[\mu_\phi(X) \mid Y \in B_i]$; the bin diameter adds a quantization error bounded by $\rho_{\max}\, \mathrm{diam}(B_i) + \rho_{\max}\, \mathrm{diam}(B_j)$ (by the upper bound of Eq. equation 5), which vanishes as the bins shrink. The separation statement itself is pointwise and needs no binning.

*Proof of Theorem 2.1.* Since $\Sigma_\theta(y) \preceq \tau_{\max}^2 I_d$, the Gaussian KL contains a mean term at least $\|\mu_\phi(x) - m_\theta(y)\|^2/(2\tau_{\max}^2)$. Hence $\mathcal{D}(\widehat{w}) \le \mathcal{K}(\widehat{w})$. For any feasible $w_0$, near-optimality and non-negativity of the prediction loss give

$$\beta\mathcal{K}(\widehat{w}) \le \mathcal{R}(\widehat{w}) + \beta\mathcal{K}(\widehat{w}) \le \mathcal{R}(w_0) + \beta\mathcal{K}(w_0) + \varepsilon.$$

Division by $\beta$ proves Eq. equation 16; substituting $\mathcal{K}(w_0) \le \delta_{\mathrm{app}}$ and $\mathcal{R}(w_0) \le C_{\mathrm{app}}$ proves Eq. equation 17. $\square$

*Proof of Proposition B.1.* For OPB let $A_Y := a q_{0,Y}$. Conditional on $X$, orthonormality gives $\mathbb{E}[A_Y \mid X] = a\sum_k \eta_k(X) q_{0,k}$ and

$$\mathbb{E}\big[\|A_Y - \mathbb{E}[A_Y \mid X]\|^2 \mid X\big] = a^2\left(1 - \|\eta(X)\|_2^2\right).$$

The conditional bias–variance identity then yields

$$\mathbb{E}\|h(X) - A_Y\|^2 = a^2\mathbb{E}[1 - \|\eta(X)\|_2^2] + \mathbb{E}\left\|h(X) - a\sum_k \eta_k(X) q_{0,k}\right\|^2.$$

Adding the diagonal-Gaussian covariance term $V_0$ proves Eq. equation 34. For EPB set $A_Y := \rho\tilde{Y}u_0$. Then $\mathbb{E}[A_Y \mid X] = \rho\nu(X)u_0$ and $\mathbb{E}[\|A_Y - \mathbb{E}[A_Y \mid X]\|^2 \mid X] = \rho^2\,\mathrm{Var}(\tilde{Y} \mid X)$. The same identity plus $V_0$ proves Eq. equation 35. $\square$

*Proof of Corollary B.1.* For classification, the assumed comparator $q(z|x) = p_\theta(z|y)$ has $\mathcal{K}(w_0) = 0$. Its Bayes cross-entropy is $C_{\mathrm{G}} = H(Y \mid Z_{\mathrm{align}}) \le \ln K$; with shared covariance the determinant factors cancel and its posterior is the anchor-energy rule. Applying Eq. equation 16 gives Eq. equation 36. For regression the aligned comparator again has zero KL. Under the tied head its error is $(\tau/\rho)\eta$, $\eta \sim \mathcal{N}(0,1)$, so its standardized squared risk is $\tau^2/\rho^2$; the same theorem gives Eq. equation 37. $\square$

*Proof of Corollary B.2.* Theorem 2.1 gives $\mathbb{E}[D(Y)] \le B_0$. For (a), $D_k \le B_0/\pi_k$; substituting this into Theorem 2.2(a) gives Eq. equation 38. For (b), Markov's inequality gives $P_Y(D(Y) > t) \le B_0/t$; on $S_t$ the pointwise inequality of Theorem 2.2(b) applies with $D(y), D(y') \le t$. $\square$

*Proof of Proposition B.2.* By Jensen's inequality the left side is at least $\frac{1}{2\tau^2}\sum_k \pi_k\|c - a q_{\theta,k}\|^2 = \frac{1}{2\tau^2}\left(\|c\|^2 - 2a\, c^\top\sum_k \pi_k q_{\theta,k} + a^2\right)$. Minimizing over $c$ gives $c = a\sum_k \pi_k q_{\theta,k}$; by orthonormality $\|\sum_k \pi_k q_{\theta,k}\|^2 = \sum_k \pi_k^2$, yielding Eq. equation 39. $\square$

*Remark* A.3 (What the framework constrains (and what it does not)). The constraints of the geometric prior bind the reference, not the deployed representation: for any orthonormal frame $\{q'_k\}$ and the class-constant posterior $q(z|Y = k) = \mathcal{N}(a\, q'_k, \tau^2 I_d)$, the trainable frame can rotate to match ($q_{\theta,k} \to q'_k$), yielding $\mathrm{KL} = 0$, $I(X; Z|Y) = 0$, and $I(Y; Z) > 0$: a full-retention label code with a zero certificate. The framework therefore constrains the geometric *form* of the label code (precisely what CEB left unspecified), and does not claim that the certificate meters retained label information. In the fixed-covariance canonical instance, the remaining freedom at certificate zero is the frame rotation. The covariance-relaxed real-data protocol has an additional but log-clamped matching channel; the comparison theorem applies with the resulting $\tau^2_{\max}$, and Section 3.4 reports that channel rather than treating it as fixed.

### A.3 CONSTRUCTION DETAILS

**Assumption A.1** (Full rank)**.** *$U_\theta$ has full column rank wherever the polar factor and its backward pass are evaluated; this is monitored rather than imposed by an extra loss.*

**Polar permutation equivariance.** For any label permutation matrix $P$,

$$Q_\theta(U_\theta P) = U_\theta P\big(P^\top U_\theta^\top U_\theta P\big)^{-1/2} = U_\theta\big(U_\theta^\top U_\theta\big)^{-1/2} P = Q_\theta(U_\theta)\, P, \tag{29}$$

since $PP^\top = I_K$ and the inverse square root conjugates under $P$. Permuting the label numbering therefore permutes the anchors accordingly, and no class index plays a distinguished role.

**Proposition A.3** (Orthogonal class-prior geometry by construction)**.** *Under Eqs. equation 7–equation 8, the class means $a_k = a\, q_{\theta,k}$ satisfy Eq. equation 9; equivalently, the class-mean Gram matrix is $a^2 I_K$: under Assumption A.1, the prior supplies a class-separated spherical code at every training step, whatever the optimizer does.*

*Proof.* Eq. equation 9 follows from $Q_\theta^\top Q_\theta = I_K$ after scaling by $a$; in particular $\|a_i - a_j\|^2 = \|a_i\|^2 + \|a_j\|^2 - 2a_i^\top a_j = 2a^2$. ☐

**Proposition A.4** (Anchor-energy classifier equivalence)**.** *The scores $s_k(z) = -\|z - a q_{\theta,k}\|^2/(2\tau^2) + \log \pi_k$ are Gaussian Bayes scores up to a class-independent term. For uniform classes, the corresponding linear weights are $(a/\tau^2) q_{\theta,k}$.*

**Initialization, monitoring, and re-initialization.** The posterior logvar head is zero-initialized ($\sigma^2 = \tau^2 = 1$ at initialization, so training starts at $\mathrm{KL} \approx \frac{1}{2\tau^2} a^2$); the prior encoder is identity-initialized, so the initial raw matrix is $U_\theta = [e_1, \dots, e_K]$, full column rank, with a stable polar-factor backward (the verified condition of Assumption A.1). The full-rank condition is *verified, not imposed*: $\sigma_{\min}(U_\theta)$ is monitored along training and stays bounded away from zero in the reported runs, and no additional regularizer is added to the objective, which remains Eq. equation 11 exactly. Near rank deficiency the polar factor's gradient is delicate; should the monitor flag a near-deficient step, the recommended handling is a re-initialization of the prior encoder (the reported runs never triggered it).

**Covariance protocol.** Equations equation 8 and equation 12 define the fixed-covariance canonical instances used for the exact Bayes-head equivalence and the controlled label-ambiguity experiment of Section 3.3. The real-data benchmark protocol retains the common CEB parameterization of a learned label-conditional diagonal covariance, with log-variance clamped to a fixed interval. It is therefore a covariance-relaxed member of $\mathcal{G}$: the OPB/EPB mean geometry and readout scale remain fixed, while Eq. equation 3 holds with the clamp-induced $\tau^2_{\max}$. Proposition A.4 is an exact Gaussian Bayes equivalence only for the shared-covariance canonical instance; in the relaxed benchmark it remains a zero-parameter readout of the same anchor means. The direct diagnostics in Section 3.4 use each checkpoint's realized conditional covariance rather than silently substituting the canonical $\tau^2 I$.

**EPB axis choices.** Three variants of the direction: (A) *fixed axis*, $u = [1, 0, \dots, 0]$, with no trainable prior direction: the theoretically cleanest ablation, at the price of prespecifying a latent coordinate; (B) *trainable normalized axis*, $u = W/\|W\|$ with $W$ trained and normalized at every step, our main variant, which keeps the scale $\rho$ exact while allowing the model to choose the axis; (C) *unconstrained linear axis*, $\mu_p(y) = W\tilde{y}$: the simplest ablation, still isometric by linearity but with a free scale that can shrink and weaken the geometry. We recommend (B), use (A) to test whether the trainable rotation carries the benefit, and keep (C) as a diagnostic. The main variant uses no bias: an affine offset cancels in pairwise distances but is a free extra degree of matching, and $\tilde{y}$ is already centered.

**EPB isometry notes.** The isometry is exact and global: any nonzero linear map of a scalar label is isometric up to its norm, and the normalization fixes that norm to $\rho$; unlike fixed random-feature anchors, whose pairwise distances grow nonlinearly and saturate, the isometric axis preserves label distances at every scale (with the price that anchor norms grow with $|\tilde{y}|$; a line and a fixed-radius sphere intersect in at most two points, so isometry and equal norms cannot both hold). All prior means lie on one line; for $y_i < y_j < y_k$ the distances add, $\|\mu_p(y_i) - \mu_p(y_k)\| = \|\mu_p(y_i) - \mu_p(y_j)\| + \|\mu_p(y_j) - \mu_p(y_k)\|$: order and magnitude of label differences are encoded exactly.

**Multidimensional regression labels.** For $y \in \mathbb{R}^m$ ($m \leq d$), the same construction extends verbatim: take $W \in \mathbb{R}^{d\times m}$, its thin polar factor $Q = W(W^\top W)^{-1/2}$, and the prior $\mu_p(y) = \rho\, Q\, \tilde{y}$, which is isometric in the label metric; the scalar case is $m = 1$ with $Q = u$. The label metric must be declared before standardization, otherwise "isometric" has no unique meaning.

### A.4 Additional results

**Lemma A.1** (OPB KL decomposition)**.** *For a diagonal posterior covariance and the prior of Eq. equation 8,*

$$\mathrm{KL}\big(q_\phi(z|x)\,\|\,p_\theta^{\mathrm{OPB}}(z|y)\big) = \frac{1}{2\tau^2}\|\mu_\phi(x) - a\, q_{\theta,y}\|^2 + \frac{1}{2}\sum_{r=1}^{d}\Big[\frac{\sigma^2_{\phi,r}(x)}{\tau^2} - 1 - \log\frac{\sigma^2_{\phi,r}(x)}{\tau^2}\Big], \quad (30)$$

*a class-anchor mismatch term plus a variance mismatch term that is non-negative and vanishes exactly at $\sigma^2_{\phi,r}(x) = \tau^2$ for all $r$. The mismatch term is quadratic in the displacement to a reference whose scale and covariance cannot move: this is the mechanism behind the strict convexity of Proposition A.5 below.*

*Proof.* Apply the diagonal-Gaussian KL formula with posterior mean $\mu_\phi(x)$, prior mean $a\, q_{\theta,y}$, posterior variances $\sigma^2_{\phi,r}(x)$, and the fixed prior variance $\tau^2$. □

**Corollary A.1** (Label-information lower bound in the aligned Gaussian model)**.** *Assume uniform classes and exact alignment $q(z|Y = k) = \mathcal{N}(a\, q_{\theta,k}, \tau^2 I_d)$. The pairwise error probability of the nearest-anchor classifier is $P_{\mathrm{pair}} = \Phi\big(-\frac{a}{\sqrt{2}\,\tau}\big)$, with $P_e \leq \min\{1, (K-1)P_{\mathrm{pair}}\}$, and whenever the displayed bound $\bar{P}_e \leq 1 - 1/K$, Fano's inequality gives $I(Y;Z) \geq \log K - h_2(\bar{P}_e) - \bar{P}_e \log(K-1)$. This links the anchor scale-to-noise ratio $a/\tau$ to the retained label information, conditionally on the aligned Gaussian model.*

*Proof.* Two anchors have distance $\sqrt{2}a$ by Proposition A.3; the equal-covariance binary decision boundary is halfway between them, giving $\Phi(-\sqrt{2}a/(2\tau))$. The union bound and Fano's inequality on $[0, 1 - 1/K]$ conclude. □

**Proposition A.5** (Joint strict convexity and unique optimum on a restricted aligned surrogate)**.** *Consider the class-symmetric aligned family: every input of class $y$ is mapped to the posterior $q_y = \mathcal{N}(s\, q_{\theta,y}, \sigma^2 I)$ ($s \geq 0$, $\sigma^2 > 0$; anchor scale $a > 0$), and the classifier is the anchor-energy classifier of Proposition A.4 (weights $w_j = c\, q_{\theta,j}$ with scale $c = a/\tau^2$ fixed by construction). With the* stochastic training code *$z \sim q_y$ (as in the actual objective, Eq. equation 11), the training cross-entropy is*

$$g(s, \sigma^2) := \mathbb{E}_\eta\Big[\ln\Big(1 + \textstyle\sum_{j\neq y} e^{-cs + c\sigma(\eta_j - \eta_y)}\Big)\Big], \qquad \eta \sim \mathcal{N}(0, I_K), \quad (31)$$

*and the training Lagrangian is*

$$L_\beta(s, \sigma^2) = g(s, \sigma^2) + \beta\Big[\tfrac{1}{2\tau^2}(s - a)^2 + \tfrac{d}{2}\big(\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2)\big)\Big]. \quad (32)$$

*Then for every $\beta > 0$, $L_\beta$ is strictly convex jointly in $(s, \sigma)$ and has a unique joint minimizer $(s^*(\beta), \sigma^*(\beta))$, interior in $s$ since $a > 0$: the stochastic training objective has a well-defined optimum, in contrast to the deterministic IB Lagrangian, whose optimum is pinned at a corner for every $\beta$ (Proposition A.1). This is a* restricted surrogate result*: it analyzes two scalar variables on the class-symmetric aligned family, with the frame, the anchor scale, and the classifier scale held fixed; the full OPB objective, with its nonlinear posterior, trainable frame, and non-convex degrees of freedom, lies outside this analysis, and the result neither establishes $\beta$-tunability of the full objective nor makes any claim on the information plane. The deterministic evaluation code ($z = \mu$) is the $\sigma \to 0$ limit, and $g \geq \ln(1 + (K-1)e^{-cs})$ by Jensen (the log-sum-exp is convex), so the statement strictly generalizes the deterministic surrogate analysis. We do* not *claim here that the map $\beta \mapsto s^*(\beta)$ is monotone or that the sweep is one-to-one: the minimizer $\sigma^*$ responds to $\beta$ through the cross-derivative $\partial^2 g/\partial s\, \partial\sigma$, and establishing the sign of $\mathrm{d}s^*/\mathrm{d}\beta$ requires the full two-dimensional stationarity analysis, which we leave open; the $\beta$-sweep is reported empirically in the experiments. The fixed-scale condition holds for the main model by construction (Proposition A.4), so the statement describes the actual training objective up to the aligned-family idealization. We do not analyze the free-classifier variant on this surrogate: under the stochastic training code, a diverging classifier scale does not produce a degenerate solution: at $\mathrm{KL} \to 0$ the posterior variance is pinned at $\sigma^2 = \tau^2 > 0$, so the class-conditional Gaussians overlap and the training cross-entropy of a free classifier diverges with its scale rather than vanishing (for $K = 2$ it contains $\mathbb{E}[\ln(1 + e^{-c(a+\tau\xi)})]$, whose integrand grows linearly in $c$ on the event $\xi < -a/\tau$, which has positive probability). An analysis of the free classifier would require an explicit classifier-norm constraint or regularizer and is left to future work.*

*Proof.* For each fixed $\eta$, the integrand of Eq. equation 31 is a log-sum-exp of functions affine in $(s, \sigma)$ (the constant 1 has slope zero and each exponential has slope $(-c,\ c(\eta_j - \eta_y))$), hence jointly convex in $(s, \sigma)$, and the expectation $g$ is jointly convex. No strictness is claimed for $g$: for $K = 2$ the integrand depends on $(s, \sigma)$ only through the single linear combination $-cs + c\sigma(\eta_2 - \eta_1)$, so its Hessian has rank one. Write the KL term as $R(s, \sigma) := \frac{1}{2\tau^2}(s-a)^2 + \frac{d}{2}\big(\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2)\big)$; its Hessian in $(s, \sigma)$ is $\nabla^2 R = \operatorname{diag}\big(\tau^{-2},\ d\,(\tau^{-2} + \sigma^{-2})\big)$, strictly positive definite for every $\sigma > 0$, so $\beta R$ is strictly convex in $(s, \sigma)$. Hence $L_\beta = g + \beta R$ is the sum of a jointly convex and a strictly convex function, and is strictly convex in $(s, \sigma)$. Since $R \to +\infty$ as $s \to \infty$, $\sigma \to 0^+$, or $\sigma \to \infty$, the sublevel sets of $L_\beta$ are compact, and the unique joint minimizer exists (by strict convexity it is unique). For $a > 0$ the minimizer is interior in $s$: at $s = 0$, $\partial L_\beta/\partial s = -c\,\mathbb{E}[p_{\mathrm{w}}] - \beta a/\tau^2 < 0$, where $p_{\mathrm{w}}$ is the error probability of the sampled code, so $s^* > 0$. □

*Remark* A.4 (The sweep is a KL–CE surrogate curve; scope and limitations). Proposition A.5 proves joint strict convexity and uniqueness of the optimum for the stochastic training objective in the KL–CE plane on the aligned family at fixed classifier scale: a restricted surrogate result, which does not establish $\beta$-tunability of the full OPB objective, and in any case not a statement on the information plane of the Caveats paper; the results of the main text establish a structured conditional prior, an exact KL geometry, and conditional separation guarantees, and they do *not* establish a mutual information upper bound tighter than CEB's: Eq. equation 13 is the usual CEB gap identity for a particular prior family. The orthogonal-frame construction fixes all pairwise anchor distances, so the framework should not claim to recover arbitrary semantic class similarities. All geometric statements are conditional on Assumption A.1: orthonormality holds exactly by the polar construction wherever the factorization exists, and the condition is verified, not imposed, along the reported runs (Section 2.1); the construction is exactly equivariant under relabeling of the classes by Eq. equation 29, so no class index plays a distinguished role. If the frame were computed from the current input batch rather than from the all-class prior matrix, the prior would become batch-dependent and Proposition 2.2 would no longer follow directly. Finally, the anchor-energy classifier of Proposition A.4 fixes the classifier scale to $a/\tau^2$ by construction, removing the independent classifier-margin escape route in the aligned surrogate of Proposition A.5; whether a free classifier (with an explicit norm constraint or regularizer) admits its own degenerate route is left to future work, and the free linear classifier is kept in the experiments as a variant for comparison. Whether the joint optimum of Proposition A.5 moves monotonically in $\beta$ is likewise open: it requires the full two-dimensional stationarity analysis, in which the optimal variance responds to $\beta$ through the cross-derivative $\partial^2 g/\partial s\, \partial\sigma$ of the stochastic cross-entropy. The trainable frame and axis constitute a rotation channel that this paper does not address: the certificate's zero set contains the full-retention orthogonal label codes at any

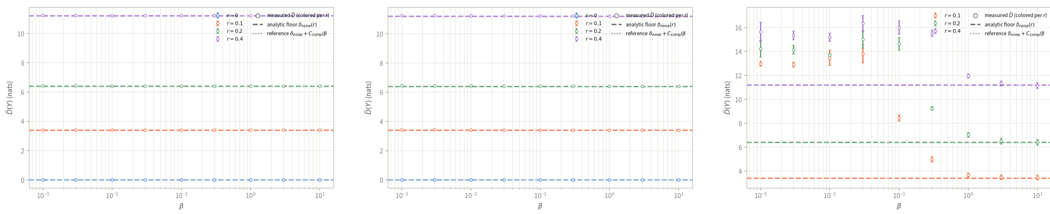

Figure 5: Sensitivity analyses for the controlled label-ambiguity experiment. (a) Posterior variance fixed to the prior variance. (b) Trainable stateless polar frame. (c) Labels sampled rather than integrated with their exact population weights; failed comparator-gate configurations are marked. Dashed lines denote the analytic population ambiguity floors. Points are means and error bars are standard deviations over five seeds.

frame rotation (the reference can rotate to meet them), and the framework pins the *geometry of the code*; it does not claim that the certificate reads off the retained label information. Quantifying how much of the anchor gap is absorbed by frame rotation rather than by posterior motion is left to future work (a frozen-frame ablation isolates it).

### A.5 Controlled label-ambiguity sensitivity analyses

Training evaluates the population objective by weighting all $K^2$ clean/observed-label pairs by the noise model, with a descending-$\beta$ warm start to remove avoidable optimization error at the small-$\beta$ end; the comparator risk $C_{\mathrm{comp}}(r)$ is estimated by Monte Carlo. The population-weighted experiment of Section 3.3 uses a fixed frame and the standard trainable posterior variance. Figure 5 changes these choices one at a time. Fixing the posterior variance to $\sigma^2 \equiv \tau^2$ leaves the measured mismatch on the analytic floors, confirming that the result is not produced by a covariance compensation channel. Replacing the fixed frame by the trainable stateless polar frame gives the same result, so the floor is invariant to the learned rigid frame rotation.

The third panel replaces the exact population weights by sampled noisy labels. It shows the expected finite-sample displacement from the population floor at weak compression and convergence toward the empirical floor as compression strengthens. Six sampled-label configurations at large $\beta$ fail the explicit comparator-objective gate (twelve stored rows because both best and final checkpoints are retained); they are marked as optimization failures rather than interpreted as violations of Eq. equation 16. The population-weighted main grid and both structural controls satisfy the gate at every point.

### A.6 Direct mismatch definitions and separation diagnostics

This section collects the diagnostic panels deferred from Section 3.4: the decoupling scatter, the separation-bound and slack readings, and the regression diagnostics, together with the finite-bin definitions. All statistics are first computed within each seed and then aggregated across the five seeds; class or bin pairs are not treated as independent replicates. Every checkpoint in both requested grids is present, and no collapse or variance-explosion point is removed.

The deferred classification panels complete the picture (Fig. 6). The decoupling scatter (a) shows MC-10 accuracy staying within 82–86$\%$ while $\widehat{D}$ spans three orders of magnitude: mechanism and task metric are decoupled. Panel (b) shows the raw bound rising toward the declared $\sqrt{2a}$ exactly where the mismatch corrections shrink, so the slack narrows where the bound is informative; CEB's mismatch is plotted descriptively, without a bound, since its reference has no declared $\Delta$. The slack percentiles (c) stay non-negative at every reported point: the bound is consistent but conservative.

**Regression.** We partition the standardized test labels into 20 equal-width bins and merge any bin containing fewer than 30 examples. Let $\bar{y}_b$ be the mean label of bin $B_b$, $\widehat{m}_b$ its posterior center, $\widehat{D}_b$ its average sample-to-own-anchor mismatch, and $h_b := \mathrm{diam}(B_b)$. The finite-bin lower bound is

$$\widehat{\mathrm{LB}}_{bc} = \rho|\bar{y}_b - \bar{y}_c| - \sqrt{2\widehat{\tau}^2_{\max}\widehat{D}_b} - \sqrt{2\widehat{\tau}^2_{\max}\widehat{D}_c} - \rho(h_b + h_c), \qquad (33)$$

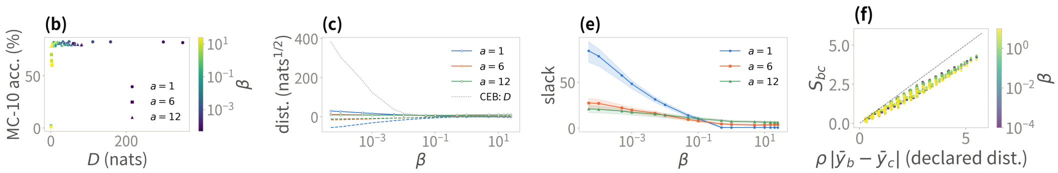


Figure 6: Direct mismatch diagnostics deferred from the main text. (a) MC-10 accuracy against $\widehat{D}$ (color $= \beta$, marker $= a$). (b) Mean separation (solid) and raw lower bound (dashed) against $\beta$; gray dotted: CEB, descriptive. (c) Slack percentiles. (d) Cal. Housing bin-pair separation against the declared prior distance at $\rho = 6$ (color $= \beta$; identity line: exact transfer). Means and standard deviations are over five seeds; no checkpoint is omitted.

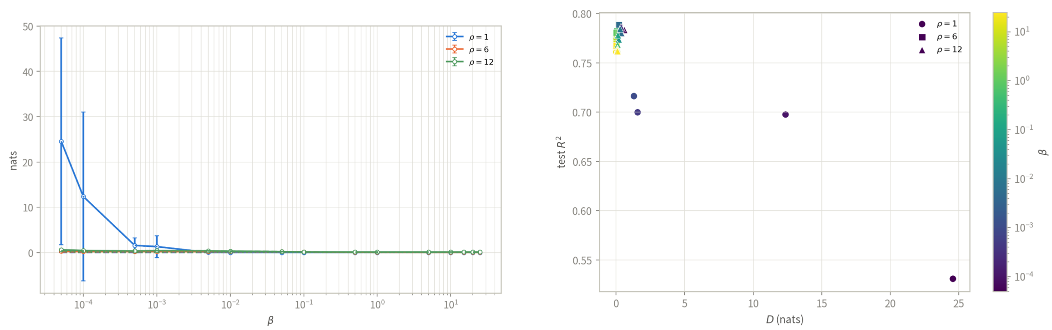


Figure 7: Cal. Housing mismatch sweep panels (deferred from the main text). (a) Anchor mismatch $\widehat{D}$ against $\beta$, by isometric scale $\rho$, decomposed into axial and off-axis contributions. (b) Test $R^2$ against $\widehat{D}$; color denotes $\beta$ and marker shape $\rho$. Means and standard deviations are over five seeds; no checkpoint is omitted.

where the last term is the quantization correction of Remark A.2. On Housing with $\rho = 6$, mismatch falls from $0.264$ to $0.026$, the axial slope remains $0.73$–$0.77$, and the mismatch is almost entirely axial (about $0.2$ against $0.004$ off-axis), while $R^2$ changes only from $0.786$ to $0.767$ over the sweep: a tenfold mismatch reduction need not imply a comparable task-metric gain. Figure 6d plots the actual bin-center distances against the declared isometric distances at $\rho = 6$: the cloud hugs a line of slope $0.73$–$0.77$ below the identity line at all four $\beta$ values, so the posterior follows the declared metric structure at about three quarters of its scale, and the scale-transfer ratio is essentially $\beta$-invariant. After the quantization correction, $43\%$ of bin pairs retain a positive bound. Figure 7 shows the two regression sweep panels: the mismatch–$\beta$ curves with their axial/off-axis decomposition (the dotted off-axis contribution stays below $0.01$ nats, so the posterior lies essentially on the declared axis), and the $R^2$-against-$\widehat{D}$ scatter whose band stays within $0.75$–$0.79$ while $\widehat{D}$ spans orders of magnitude.

**Normalization and variance extremes.** The empirical mismatch uses one maximum conditional-reference variance per checkpoint, whereas the exact Gaussian KL uses every sample's and coordinate's own variance. They therefore need not coincide: averaged over the ImageNet-100 grid, $\widehat{D} = 41.2$, the KL mean term is $46.5$, and the KL variance term is $51.4$. On Housing, a small number of weak-compression, large-$\rho$ variance explosions raise the across-grid mean KL variance term to $2.8 \times 10^6$. These checkpoints remain in all plots. Because their $\widehat{\tau}^2_{\max}$ is also large, the normalized $\widehat{D}$ can fall even as the KL variance term grows; the unnormalized axial/off-axis components and total KL are therefore reported alongside it.

Table 5: Test accuracy (%) and test $R^2$ with standard deviations over runs: the plain backbone, the variational baselines, and NCBD (classification settings only); same configurations as Table 1.

| Dataset | Backbone | Base | VIB | SVIB | NIB | NCBD |
|---|---|---|---|---|---|---|
| MNIST | MLP | 98.33±0.12 | 98.28±0.11 | 98.15±0.13 | 98.27±0.07 | 97.95±0.23 |
| MNIST | CNN | 99.13±0.05 | 99.28±0.09 | 99.23±0.03 | 99.22±0.06 | 99.18±0.07 |
| ImageNet-100 | MLP | 83.12±0.14 | 82.84±0.44 | 83.02±0.37 | 82.35±0.27 | 83.64±0.07 |
| ImageNet-100 | CNN | 83.40±0.13 | 82.94±0.37 | 83.11±0.86 | 82.91±0.35 | 80.56±1.86 |
| Cora | GCN | 87.68±0.46 | 86.83±0.28 | 86.46±1.87 | 86.35±0.68 | 87.34±1.60 |
| IMDb | RNN | 84.85±0.68 | 85.50±0.91 | 85.99±0.30 | 86.14±0.92 | 86.69±0.52 |
| AG News | RNN | **92.94±0.15** | 92.72±0.17 | 92.86±0.21 | 92.91±0.22 | 92.37±0.37 |
| Cal. Housing | MLP | 0.7618±0.0238 | $\underline{0.7901\pm0.0049}$ | 0.7897±0.0058 | 0.7862±0.0035 | – |
| STS-B | RNN | **0.2899±0.0113** | 0.2719±0.0164 | 0.2697±0.0202 | 0.2417±0.0221 | – |
| ZINC | GCN | $\underline{0.6348\pm0.0044}$ | 0.6339±0.0033 | 0.6325±0.0026 | 0.6178±0.0099 | – |
| AgeDB | MLP | 0.2180±0.0409 | 0.3173±0.0101 | 0.3385±0.0286 | 0.2710±0.0235 | – |
| AgeDB | CNN | 0.3914±0.0194 | 0.3342±0.1894 | 0.3323±0.1870 | 0.3337±0.1889 | – |

Table 6: Test accuracy (%) and test $R^2$ with standard deviations over runs: CEB, DVCCA, AdaCap (regression settings only), and our method; same configurations as Table 1. GPB and GPB-L are separated from the references by the vertical rule.

| Dataset | Backbone | CEB | DVCCA | AdaCap | GPB | GPB-L |
|---|---|---|---|---|---|---|
| MNIST | MLP | $\underline{98.43\pm0.08}$ | 98.24±0.17 | – | **98.54±0.09** | $\underline{98.43\pm0.07}$ |
| MNIST | CNN | 99.30±0.05 | 99.29±0.10 | – | **99.32±0.05** | $\underline{99.31\pm0.05}$ |
| ImageNet-100 | MLP | 82.77±0.22 | 82.99±0.43 | – | **84.40±0.21** | $\underline{84.01\pm0.23}$ |
| ImageNet-100 | CNN | 82.97±0.67 | 83.21±0.70 | – | **86.08±0.21** | $\underline{84.88\pm0.80}$ |
| Cora | GCN | 86.27±0.38 | 86.90±0.94 | – | **88.12±0.55** | $\underline{87.86\pm0.87}$ |
| IMDb | RNN | 85.55±0.73 | 85.69±0.23 | – | $\underline{86.84\pm0.57}$ | **86.99±0.35** |
| AG News | RNN | 92.83±0.11 | 92.78±0.15 | – | 92.87±0.12 | $\underline{92.92\pm0.14}$ |
| Cal. Housing | MLP | 0.7894±0.0018 | 0.7878±0.0048 | 0.7758±0.0023 | 0.7895±0.0051 | **0.7914±0.0034** |
| STS-B | RNN | 0.2668±0.0127 | 0.2713±0.0218 | 0.1613±0.0188 | $\underline{0.2762\pm0.0152}$ | 0.2755±0.0134 |
| ZINC | GCN | 0.6257±0.0044 | 0.6294±0.0026 | **0.6770±0.0019** | 0.6344±0.0031 | 0.6323±0.0016 |
| AgeDB | MLP | 0.3317±0.0251 | 0.3180±0.0228 | $\underline{0.3482\pm0.0209}$ | **0.3536±0.0076** | 0.3320±0.0080 |
| AgeDB | CNN | 0.3882±0.0275 | 0.3205±0.1811 | $\underline{0.4684\pm0.0094}$ | **0.4700±0.0072** | 0.4388±0.0220 |

### A.7 MAIN RESULTS WITH STANDARD DEVIATIONS

Tables 5 and 6 report the same configurations as the main table with standard deviations over runs, split by column groups for width (the baselines table includes NCBD and the variants table includes AdaCap); bold and underline follow the same mean-based rule as Table 1.

### A.8 PAIRED-DIFFERENCE CONFIDENCE INTERVALS

Tables 7–10 report, for each setting, the paired difference of GPB over every baseline (GPB minus baseline) with a bootstrap $95\%$ confidence interval: the differences are paired across the shared seeds, the interval is the percentile interval of $B = 10{,}000$ resamples of the per-run differences (fixed seed, reproducible), and the non-inferiority margin is $\delta = 0.2$ percentage points of accuracy or $\delta = 0.005$ in $R^2$. Bold marks a lower bound above zero (significant advantage) and $^{\dagger}$ a lower bound within the margin (tied); the best configuration per method is the same as in the main table. The last table covers the external non-IB baselines, NCBD on the classification settings and AdaCap on the regression settings; the only interval lying entirely below zero is AdaCap on ZINC ($-0.043$ $R^2$).

### A.9 INFORMATION-PLANE PANELS FOR CAL. HOUSING

Figure 8 completes panels (c)–(d) of Fig. 3 in the main text with the regression setting, under the same $\beta$ grid, axis conventions, and five runs per point; here $I(Y;Z)$ is the Gaussian approximation $\frac{1}{2}\ln\bigl(\mathrm{Var}(y)/\mathrm{Var}(y-\hat{y})\bigr)$, a heuristic rather than a bound. On Cal. Housing this approximation

Table 7: Paired-difference 95% confidence intervals of GPB over the plain backbone and VIB (GPB minus baseline, per setting; classification in percentage points, regression in $R^2$). Bold: lower CI bound above zero; †: lower bound within the non-inferiority margin $\delta$ (tied).

| Dataset | Backbone | Base | VIB |
|---|---|---|---|
| MNIST | MLP | **+0.2 [+0.2, +0.3]** | **+0.3 [+0.2, +0.3]** |
| MNIST | CNN | **+0.2 [+0.2, +0.2]** | +0.0 [-0.0, +0.1]† |
| ImageNet-100 | MLP | **+1.3 [+1.2, +1.4]** | **+1.6 [+1.4, +1.9]** |
| ImageNet-100 | CNN | **+2.7 [+2.5, +2.9]** | **+3.1 [+3.0, +3.4]** |
| Cora | GCN | +0.4 [-0.1, +1.1]† | **+1.3 [+0.7, +1.7]** |
| IMDb | RNN | **+2.0 [+1.2, +2.7]** | **+1.3 [+0.6, +2.3]** |
| AG News | RNN | -0.1 [-0.2, +0.0]† | **+0.2 [+0.1, +0.2]** |
| Cal. Housing | MLP | **+0.028 [+0.008, +0.048]** | -0.001 [-0.005, +0.003]† |
| STS-B | RNN | -0.014 [-0.022, -0.006] | +0.004 [-0.017, +0.022] |
| ZINC | GCN | -0.000 [-0.003, +0.002]† | +0.001 [-0.004, +0.004]† |
| AgeDB | MLP | **+0.136 [+0.104, +0.167]** | **+0.036 [+0.026, +0.046]** |
| AgeDB | CNN | **+0.079 [+0.066, +0.091]** | **+0.136 [+0.043, +0.302]** |
| | | 8*/3† | 8*/3† |

Table 8: Paired-difference 95% confidence intervals of GPB over SVIB and NIB (GPB minus baseline, per setting; classification in percentage points, regression in $R^2$). Bold/† as in the baselines table.

| Dataset | Backbone | SVIB | NIB |
|---|---|---|---|
| MNIST | MLP | **+0.4 [+0.3, +0.5]** | **+0.3 [+0.2, +0.3]** |
| MNIST | CNN | **+0.1 [+0.1, +0.1]** | **+0.1 [+0.1, +0.2]** |
| ImageNet-100 | MLP | **+1.4 [+1.1, +1.7]** | **+2.1 [+1.8, +2.2]** |
| ImageNet-100 | CNN | **+3.0 [+2.4, +3.9]** | **+3.2 [+2.9, +3.5]** |
| Cora | GCN | **+1.7 [+0.2, +3.4]** | **+1.8 [+1.1, +2.4]** |
| IMDb | RNN | **+0.8 [+0.2, +1.4]** | **+0.7 [+0.2, +1.2]** |
| AG News | RNN | +0.0 [-0.1, +0.1]† | -0.0 [-0.2, +0.1]† |
| Cal. Housing | MLP | -0.000 [-0.006, +0.004] | +0.003 [-0.001, +0.009]† |
| STS-B | RNN | +0.006 [-0.021, +0.028] | **+0.035 [+0.011, +0.058]** |
| ZINC | GCN | +0.002 [-0.001, +0.004]† | **+0.017 [+0.010, +0.026]** |
| AgeDB | MLP | +0.015 [-0.010, +0.041] | **+0.083 [+0.062, +0.100]** |
| AgeDB | CNN | **+0.138 [+0.045, +0.303]** | **+0.136 [+0.040, +0.302]** |
| | | 7*/2† | 10*/2† |

consistently reads above the InfoNCE bound, so the certificate panel reflects the estimator gap rather than the geometry; the information plane remains informative: CEB compresses itself off the task ($R^2 = 0.49$ at $\beta = 0.1$, $0.18$ at $\beta = 0.5$) while every EPB curve keeps $R^2 \geq 0.76$ flat out to $\beta = 25$.

## B Additional theory and experiments

This appendix collects the detailed material omitted from the nine-page main text. All reported numbers use the same checkpoints and seeds as the main comparison.

### B.1 Deferred theory statements

**Proposition B.1** (Ambiguity–approximation decomposition). *For OPB classification with $\eta_k(x) = P(Y = k \mid X = x)$ and a comparator mean $h(x)$,*

$$\mathcal{K}(w_0) = \frac{a^2}{2\tau^2}\mathbb{E}[1 - \|\eta(X)\|_2^2] + \frac{1}{2\tau^2}\mathbb{E}\left\|h(X) - a\sum_k \eta_k(X)q_{0,k}\right\|^2 + V_0, \tag{34}$$

*where $V_0$ is the covariance mismatch. For EPB regression,*

$$\mathcal{K}(w_0) = \frac{\rho^2}{2\tau^2}\mathbb{E}[\mathrm{Var}(\tilde{Y} \mid X)] + \frac{1}{2\tau^2}\mathbb{E}\|h(X) - \rho\nu(X)u_0\|^2 + V_0. \tag{35}$$

Table 9: Paired-difference 95% confidence intervals of GPB over CEB, DVCCA, and the free-head ablation GPB-L (GPB minus baseline, per setting; classification in percentage points, regression in $R^2$). Bold/† as in the baselines table.

| Dataset | Backbone | CEB | DVCCA | GPB-L |
|---|---|---|---|---|
| MNIST | MLP | **+0.1 [+0.0, +0.2]** | **+0.3 [+0.2, +0.4]** | **+0.1 [+0.0, +0.2]** |
| MNIST | CNN | +0.0 [-0.0, +0.1]† | +0.0 [-0.1, +0.1]† | +0.0 [-0.0, +0.1]† |
| ImageNet-100 | MLP | **+1.6 [+1.5, +1.7]** | **+1.4 [+1.1, +1.7]** | **+0.4 [+0.2, +0.5]** |
| ImageNet-100 | CNN | **+3.1 [+2.4, +3.7]** | **+2.9 [+2.4, +3.5]** | **+1.2 [+0.5, +1.9]** |
| Cora | GCN | **+1.8 [+1.5, +2.3]** | **+1.2 [+0.4, +1.9]** | +0.3 [-0.5, +1.3] |
| IMDb | RNN | **+1.3 [+0.3, +2.1]** | **+1.1 [+0.8, +1.5]** | -0.2 [-0.6, +0.5] |
| AG News | RNN | +0.0 [-0.1, +0.2]† | +0.1 [-0.0, +0.2]† | -0.0 [-0.2, +0.1]† |
| Cal. Housing | MLP | +0.000 [-0.003, +0.004]† | +0.002 [-0.005, +0.009] | -0.002 [-0.005, +0.002] |
| STS-B | RNN | +0.009 [-0.009, +0.025] | +0.005 [-0.020, +0.026] | +0.001 [-0.019, +0.017] |
| ZINC | GCN | **+0.009 [+0.005, +0.011]** | **+0.005 [+0.002, +0.007]** | +0.002 [-0.001, +0.005]† |
| AgeDB | MLP | **+0.022 [+0.006, +0.038]** | **+0.036 [+0.012, +0.057]** | **+0.022 [+0.019, +0.025]** |
| AgeDB | CNN | **+0.082 [+0.064, +0.099]** | **+0.149 [+0.065, +0.310]** | **+0.031 [+0.018, +0.046]** |
| | | 8*/3† | 8*/2† | 5*/3† |

Table 10: Paired-difference 95% confidence intervals of GPB over the external non-IB baselines NCBD (classification settings only) and AdaCap (regression settings only); GPB minus baseline, classification in percentage points, regression in $R^2$. Bold/† as in the baselines table.

| Dataset | Backbone | NCBD | AdaCap |
|---|---|---|---|
| MNIST | MLP | **+0.6 [+0.4, +0.8]** | – |
| MNIST | CNN | **+0.1 [+0.1, +0.2]** | – |
| ImageNet-100 | MLP | **+0.8 [+0.6, +0.9]** | – |
| ImageNet-100 | CNN | **+5.5 [+4.3, +6.9]** | – |
| Cora | GCN | **+0.8 [+0.0, +1.9]** | – |
| IMDb | RNN | +0.2 [-0.5, +0.7] | – |
| AG News | RNN | **+0.5 [+0.2, +0.8]** | – |
| Cal. Housing | MLP | – | **+0.014 [+0.009, +0.019]** |
| STS-B | RNN | – | **+0.115 [+0.086, +0.135]** |
| ZINC | GCN | – | -0.043 [-0.046, -0.041] |
| AgeDB | MLP | – | +0.005 [-0.007, +0.021] |
| AgeDB | CNN | – | +0.002 [-0.004, +0.006]† |
| | | 6*/0† | 2*/1† |

**Corollary B.1** (Zero-floor exact-realizability special cases)**.** *Under deterministic labels, exact posterior–prior matching, and a compatible Bayes head, $\mathcal{K}(w_0) = 0$ and Theorem 2.1 gives*

$$\mathcal{D}(\widehat{w}) \leq (\ln K + \varepsilon)/\beta \tag{36}$$

*for classification. For EPB regression,*

$$\mathcal{D}(\widehat{w}) \leq (\tau^2/\rho^2 + \varepsilon)/\beta. \tag{37}$$

**Corollary B.2** (Comparator-dependent separation at near-optimal solutions)**.** *Writing $B_0 = \mathcal{K}(w_0) + (\mathcal{R}(w_0) + \varepsilon)/\beta$, classification satisfies*

$$\|m_i - m_j\| \geq \|m_\theta(i) - m_\theta(j)\| - \sqrt{2\tau_{\max}^2 B_0/\pi_i} - \sqrt{2\tau_{\max}^2 B_0/\pi_j}, \tag{38}$$

*and the continuous-label version holds on $S_t = \{y : D(y) \leq t\}$ with measure at least $1 - B_0/t$ and loss $2\sqrt{2\tau_{\max}^2 t}$ in the separation bound.*

**Proposition B.2** (Positive rate cost of class-independent means)**.** *If $\mathbb{E}[\mu_\phi(X) \mid Y = k] = c$ for every class, then*

$$\mathbb{E}\left[\frac{\|\mu_\phi(X) - aq_{\theta,Y}\|^2}{2\tau^2}\right] \geq \frac{a^2}{2\tau^2}\left(1 - \sum_k \pi_k^2\right). \tag{39}$$

**Corollary B.3** (The orthogonal instance instantiates conditional posterior separation)**.** *For OPB, $\tau_{\max} = \tau$ and $\Delta = \sqrt{2}a$, so*

$$\|m_i - m_j\| \geq \sqrt{2}a - \sqrt{2\tau^2 D_i} - \sqrt{2\tau^2 D_j}. \tag{40}$$

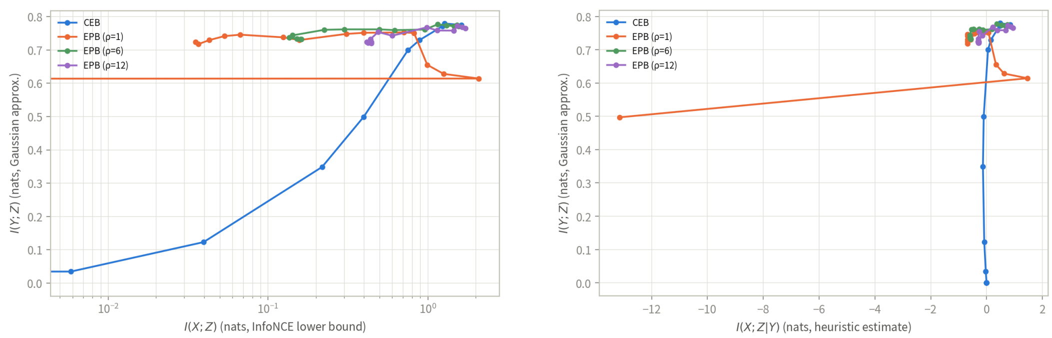


Figure 8: Information plane (a) and certificate plane (b) for Cal. Housing, companion panels to Fig. 3(c,d): identical estimators, $\beta$ grid (log $x$-scale in (a); linear where the estimated residual turns negative in (b)), five runs per point, lines connect $\beta$ in ascending order. CEB one curve per panel; EPB three ($\rho \in \{1, 6, 12\}$).

| factor | comparison | MNIST | ImageNet-100 | Housing |
|---|---|---|---|---|
| readout | GPB − GPB-L | $+0.02\,[-0.05, 0.08]$ | $+0.82\,[0.34, 1.17]$ | – |
| readout (EPB) | EPB − EPB-L | – | – | $-0.00\,[-0.01, -0.00]$ |
| readout (free geometry) | CEB-energy − CEB | $-0.21\,[-0.30, -0.12]$ | $-1.19\,[-1.50, -0.96]$ | – |
| geometry | OPB-FS − CEB-energy | $+0.32\,[0.26, 0.40]$ | $+2.67\,[2.45, 2.92]$ | – |
| geometry (EPB) | EPB-FS − CEB-tied | – | – | $+0.01\,[-0.00, 0.01]$ |
| fixed scale | GPB − OPB-FS | $-0.09\,[-0.16, -0.02]$ | $-0.04\,[-0.07, -0.01]$ | – |
| fixed scale (EPB) | EPB − EPB-FS | – | – | $+0.00\,[-0.01, 0.01]$ |
| IB increment* | GPB − NCM-ortho | $+0.34\,[0.22, 0.45]$ | $+1.89\,[1.41, 2.36]$ | – |
| prototype structure | NCM-ortho − NCM-learn | $-0.06\,[-0.20, 0.11]$ | $+0.16\,[-0.49, 0.82]$ | – |
| prototype readout alone | NCM-learn − Base | $-0.16\,[-0.28, -0.05]$ | $-0.96\,[-1.30, -0.69]$ | – |

Table 11: Factor-wise paired differences (accuracy points on MNIST and ImageNet-100, $R^2$ on Housing; 95% bootstrap percentile CIs, $B = 10{,}000$, fixed seed, paired across the shared five seeds at each variant's best configuration; positive values favor the first term). *NCM-ortho fixes the frame at $a\,e_k$ while GPB trains it, so the IB comparison additionally contains frame trainability.

## B.2 Factor-wise ablations

The factor-wise ablation ladder switches geometry, anchor scale, readout, and the IB term separately. The complete means appear in Table 3 of the main text; the paired bootstrap intervals are reported below. Table 11 reports the paired bootstrap intervals for all contrasts; the factor ranking and the sign-flip and negative-result patterns are read in the main text (Section 3.6).

## B.3 Regression compression curves

The main text shows the classification compression sweep. This companion figure gives the complete California Housing regression sweep under the same $\beta$ grid, including the weak-compression variance explosion and the strong-compression EPB plateau. The curves are diagnostics of achieved KL and task performance, not tests of the alignment rate.

## B.4 Information-plane estimates

Estimator protocol. $I(X; Z)$ is estimated by the InfoNCE lower bound (van den Oord et al., 2018) ($\log B - \mathcal{L}_{\mathrm{NCE}}$, $B = 4096$; one critic per run, trained on the training split, the bound read off the test split with $z \sim q(z|x)$); $I(Y; Z)$ by the classification lower bound $H(Y) - \mathrm{CE}$; and the residual $I(X; Z|Y)$ is the difference of the two, a heuristic rather than a bound. The true residual is nonnegative by the data-processing inequality; negative plotted values mean the estimators, not the theory, gave way. Because negative residuals occur (OPB $a = 1$; the collapsed CEB trajectory), the certificate-plane axis is drawn linear. The MNIST panels appear in Fig. 3(c,d) of the main text.

Matched-compression performance. Table 12 pairs each CEB point with the OPB configuration of nearest estimated $I(X; Z)$. At $\approx 2.0$ nat, CEB ($\beta = 0.5$) retains 86.0% while OPB ($a = 1, \beta = 0.5$)

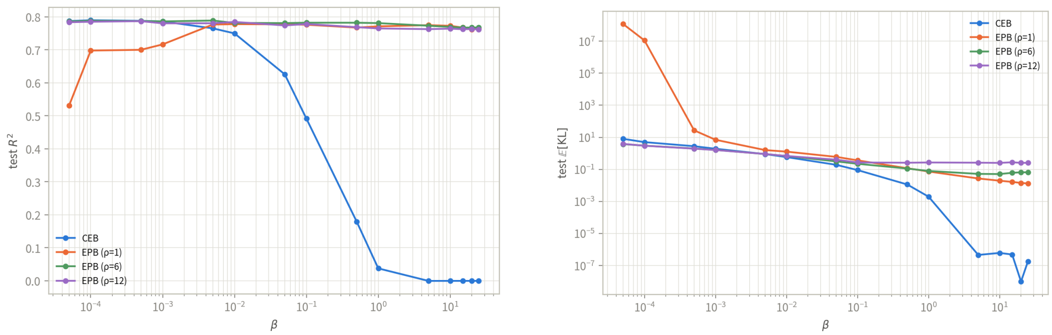


Figure 9: California Housing compression sweep (MLP). Left: test $R^2$; right: realized conditional KL over $\beta$. EPB maintains a task plateau through strong compression, while the $\rho = 1$ weak-compression points exhibit a large variance term.

| Setting | CEB $I(X;Z)$ | CEB Acc. | OPB config. | OPB $I(X;Z)$ | OPB Acc. |
|---|---|---|---|---|---|
| MNIST, $\beta = 0.1$ | 2.53 | 98.2 | $a = 1, \beta = 0.1$ | 2.55 | 98.1 |
| MNIST, $\beta = 0.5$ | 1.96 | 86.0 | $a = 1, \beta = 0.5$ | 2.01 | 97.8 |

Table 12: Matched-compression information-plane comparison (means; standard deviations are in the stored evaluation report).

retains 97.8%; at $\approx 2.5$ nat ($\beta = 0.1$) the two tie (98.2% vs. 98.1%). These are paired empirical comparisons at two estimated compression levels, not a causal attribution to geometry.

Estimator caveat. OPB ($a = 1$) at $\beta \geq 5$ must be read with care. On the deterministic $\mu$ path the accuracy stays 97.6–97.8%, for which Fano's inequality requires $I(Y;Z) \geq H(Y) - h(0.023) - 0.023 \ln 9 \approx 2.1$ nat, so the plotted $I(Y;Z)$ of 0.35–0.74 nat is a bound failure on that path: overconfident errors inflate the cross-entropy, and near-duplicate positives saturate the InfoNCE objective. On the sampled posterior the degradation is partly real: at posterior variance $\tau^2 = 1$ the radius-1 anchors cannot separate classes, and the stochastic-path clean accuracy falls to 0.84 at $\beta = 5$ and 0.59 at $\beta = 25$, against which the CE-based estimate still sits below the Fano floor. The leftward spur therefore mixes a real sampled-path collapse with an estimator failure of the deterministic path; the $a = 6, 12$ readings above are unaffected.

### B.5 Adversarial robustness details

Targeted PGD uses 20 steps, EOT-10 gradients, 10-sample MC predictions, and budgets $\varepsilon_\infty = 0.3$ and $\varepsilon_2 = 6$. Curves are shown only when clean stochastic accuracy exceeds 90%. At maximum compression, OPB ($a = 6, 12$) reaches 31.1%/30.5% targeted success under $L_\infty$, while transfer attacks from CEB to OPB reach at most 2.0%/3.4% under $L_\infty/L_2$, far below OPB's white-box attack. The transfer result is a gradient-masking check rather than a causal geometry test.

## C Related work

**Information bottleneck and conditional compression.** The information bottleneck (IB) principle formulates representation learning as retaining information about a target while discarding information about the input (Tishby et al., 2000). Deep VIB turns this principle into a neural objective by using a reparameterized stochastic encoder and a variational marginal prior (Alemi et al., 2017). Nonlinear and non-parametric bottleneck variants study alternative variational bounds and distance-based estimates of the information in a representation (Kolchinsky et al., 2019). The deterministic-scenario analysis of Kolchinsky & Tracey (2019) further shows that the usual IB multiplier can fail to sweep an interior information–accuracy trade-off when labels are deterministic; the squared-compression baseline in our experiments follows this line of work.

The conditional entropy bottleneck (CEB) replaces the marginal reference of VIB by a label-conditional reference and thereby targets conditional residual information (Fischer, 2020). Com-

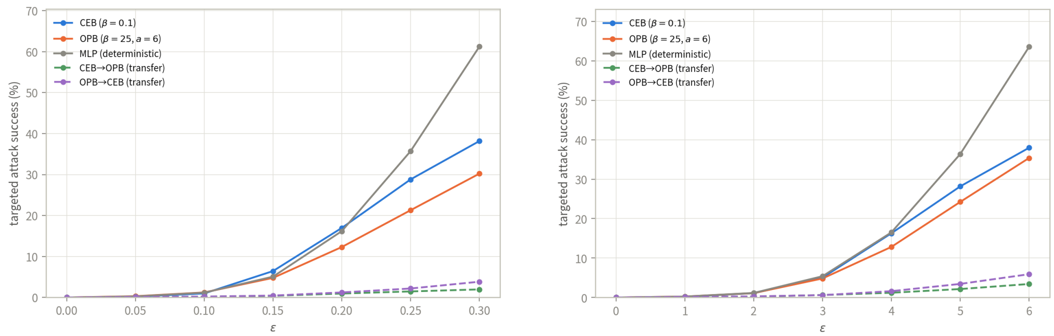


Figure 10: Transfer PGD curves on MNIST (gradient-masking check). Targeted adversaries are generated on one model and evaluated on another; CEB-to-OPB transfer remains far below OPB's own white-box attack. The targeted white-box curves are in the main text.

parisons of variational bounds clarify when this conditional reference is tighter than a marginal one (Geiger & Fischer, 2020). CEB has also been applied to large-scale visual representations and robustness evaluations (Fischer & Alemi, 2020; Lee et al., 2021), while multivariate extensions retain the same variational bottleneck viewpoint (Abdelaleem et al., 2025). These methods leave the relationships among distinct conditional reference means unspecified. GPB keeps the CEB posterior and its single conditional KL, but restricts the reference family itself: OPB declares a class frame and EPB declares a label-distance-preserving axis.

**Information bottleneck for robust representations.** Several works use a bottleneck to suppress nuisance or non-robust information. The HSIC bottleneck adds layer-wise dependence penalties and analyzes adversarial robustness (Wang et al., 2021). Adversarial Information Bottleneck explicitly incorporates adversarial objectives (Zhai & Zhang, 2022), and an NLP-specific variant separates task-relevant robust features from manipulable features (Zhang et al., 2022). Causal Information Bottleneck uses a causal decomposition for the same robust-feature goal (Hua et al., 2022). These approaches modify the information criterion, feature selection, or training examples. In contrast, the geometric constraint in GPB is placed on the conditional prior and is evaluated with the ordinary CEB KL; the robustness experiments therefore test whether a declared prior geometry is transferred to the posterior rather than introducing adversarial training as an additional objective.

**Geometric classification representations.** Neural-collapse studies document an approximately simplex-ETF arrangement of class means and classifier directions near the terminal phase of training (Papyan et al., 2020; Zhu et al., 2021). The connection between neural collapse, supervised contrastive learning, and an information bottleneck has also been analyzed explicitly (Wang & Palmer, 2023). Other works impose related geometry directly: orthogonal classifier weights can improve adversarial robustness (Xu et al., 2022); ETF classifiers have been used for early exit (Ji et al., 2023); and fixed or hierarchy-aware frames have been used to reduce classifier bias or mistake severity (Tong & Davis, 2023). Neural Collapse by Design trains features and prototypes jointly on the unit hypersphere so that the global optimum of the NONL contrastive objective is exactly the simplex ETF (Koromilas et al., 2026); we include it as a classification-only comparison baseline. Equidistant prototype embeddings provide an earlier prototype-based classification example (Deng et al., 2014).

These works constrain features, classifier weights, or prototypes through a classification loss or a fixed readout. OPB addresses a different object: the label-conditional Gaussian prior in the bottleneck. Its polar/QR frame is recomputed from the prior encoder, and the anchor-energy classifier reads the same frame without introducing an independent classifier scale. Thus the geometric class code and the conditional KL are coupled during training.

Table 13 summarizes this object-level distinction. The geometric primitive is shared with several preceding lines of work, but its role in the objective is different: OPB anchors are constrained class centers that serve simultaneously as Gaussian-prior means, KL targets, and readout anchors. In the shared-covariance canonical instance, the induced readout is an orthogonal classifier as a consequence of the Gaussian Bayes rule. It is not an ETF construction: OPB has Gram matrix $a^2 I_K$

| Approach | Constrained object | Enters an IB KL? | Readout tied to geometry? | Stochastic posterior? |
| --- | --- | --- | --- | --- |
| Prototype methods (Deng et al., 2014) | prototypes or features | No | Usually no/partial | Usually no |
| ETF / orthogonal classifiers (Xu et al., 2022; Ji et al., 2023) | classifier weights or feature directions | No | Yes | Usually no |
| Fixed or hierarchy-aware frames (Tong & Davis, 2023) | feature coordinates or a frame | No | Method-dependent | Usually no |
| CEB (Fischer, 2020) | label-conditional reference | Yes (conditional KL) | No (free head) | Yes |
| GPB (OPB / EPB) | label-conditional Gaussian prior means and covariance | Yes (conditional KL) | Yes (tied rule) | Yes |

Table 13: Conceptual boundary between geometric representation methods and GPB. The entries describe the object constrained by the method, not whether a particular implementation may contain additional components. GPB reuses familiar geometric primitives while placing them in the conditional prior used by the bottleneck and using the same prior geometry for prediction.

(zero pairwise inner products), whereas a centered simplex ETF has constant negative off-diagonal inner products in a $(K-1)$-dimensional subspace. The frame is recomputed from the prior encoder and may rotate, rather than fixing a coordinate frame or optimizing a separate prototype table.

**Distance-preserving regression and geometric priors.** Isometric representation learning has traditionally focused on preserving distances on a data manifold, including deep extensions of isomap-style mappings (Pai et al., 2022). Orthogonal label regression has also been used as a discriminative device in domain adaptation (Luo et al., 2023). These methods motivate using a declared label metric, but they generally constrain a deterministic feature map rather than the mean map of a conditional Gaussian information-bottleneck prior. EPB uses the scalar label case: after standardization, prior means lie on a normalized axis with a fixed scale, so latent prior distances equal the declared label distances up to that scale. The same axis is also the tied regression readout and the target of the conditional KL. Thus EPB is a conditional-prior use of distance geometry, not a new distance-preserving primitive. The regression information-bottleneck literature, including divergence-based bounds, provides complementary compression objectives rather than this geometric prior construction (Yu et al., 2024). AdaCap pairs a permutation-based contrastive loss with a Tikhonov closed-form output layer for small-sample tabular regression (Belucci et al., 2026); we include it as a regression-only comparison baseline.

**Geometry-aware information bottlenecks.** Recent work has used the phrase geometric information bottleneck in several distinct settings. A variational recommender learns a geometric prior for explanation generation (Yan et al., 2024); a data-scarcity formulation penalizes latent curvature and intrinsic dimension (Katende, 2025); and activation steering uses orthogonal subspaces to control language-model interventions (Doan et al., 2026). A concurrent analysis studies the relationship between information compression and class-wise geometric compression in CEB networks (Adilova et al., 2026). These studies make geometry an important representation concern, but they do not use the same conditional Gaussian prior for both compression and prediction, nor do they provide the OPB/EPB alignment-to- separation construction considered here.